MDS211 Nervous System — Past Paper Recreation

Lectures 17–29 (Exam 2 / FA-SA2) · 225 Questions · Single Best Answer

Compiled from the MDS211 past-paper archive spanning multiple cohorts (MED33–VCKMD36), covering Lectures 17–29 (Exam 2 / FA-SA2, incl. Clinical Correlation). All 225 recorded questions are included. The answer key has been fully re-verified against source neuroscience material rather than the original automated colour-detection extraction — 90 answers that were factually wrong have been corrected (a 40% error rate in the original key), and 19 remain flagged where the question is genuinely ambiguous, references an image/table not captured in the text, or has more than one defensible answer.

Lecture 17 — Motor System (27 questions)

1. About myotatic reflex of the knee, which is correct?

A. Afferent = Ib fiber
B. Efferent = Ia fiber
C. Antagonist = quadriceps
D. Heteronymous = hamstring
E. Receptor = Golgi tendon organ

2. When dynamic γ-motor neurons activate with α-motor neurons:

A. Ia inhibition occurs
B. Clonus occurs
C. Muscle won’t contract
D. Fewer Ia impulses than α alone
E. More Ia impulses than α alone

3. Which statement about muscle spindles is correct?

A. Bag: static; Chain: static
B. Bag: static & dynamic; Chain: static
C. Bag: dynamic; Chain: dynamic
D. Bag: static; Chain: static & dynamic
E. Bag: static & dynamic; Chain: dynamic

4. Regarding the knee myotatic reflex, which statement is correct?

A. Afferent = Ib fiber
B. Efferent = Ia fiber 9
C. Antagonist = quadriceps
D. Heteronymous = hamstring
E. Receptor = Golgi tendon organ

5. Regarding muscle spindles, which statement is correct?

A. Bag = static; Chain = static
B. Bag = static & dynamic; Chain = static
C. Bag = dynamic; Chain = dynamic
D. Bag = static; Chain = static & dynamic
E. Bag = static & dynamic; Chain = dynamic

6. When dynamic γ-motor neurons are activated with α-motor neurons:

A. Ia inhibition occurs
B. Clonus occurs
C. Muscle won't contract
D. Fewer Ia impulses are generated than with α-activation alone
E. More Ia impulses are generated than with α-activation alone

7. Which of the following structures is NOT part of the motor system?

A. Cerebellum
B. Basal ganglia
C. Extrapyramidal tracts
D. Cingulate gyrus of cerebrum
E. Neuromuscular junction & extrafusal muscle

8. In the biceps reflex, the antagonist muscle is the:

A. Afferent = Ib
B. Antagonist = triceps
C. Homonymous = hamstring
D. Heteronymous = biceps
E. Receptor = Golgi tendon organ

9. Which statement about muscle spindles is correct?

A. Bag: static, Chain: static
B. Bag: static & dynamic, Chain: static
C. Bag: dynamic, Chain: dynamic
D. Bag: static, Chain: static & dynamic
E. Bag: static & dynamic, Chain: dynamic

10. Damage to which structure gives lower motor neuron (LMN) signs?

A. Crus cerebri
B. Precentral gyrus
C. Motor nuclei of brainstem
D. Posterior limb of internal capsule
E. Cerebellum

11. A patient has left lower facial weakness and the tongue deviates to the left when protruded. The lesion is in the:

A. Right corticospinal tract
B. Right corticobulbar tract
C. Left facial motor nucleus + hypoglossal nucleus
D. Left corticobulbar tract + left hypoglossal nucleus
E. Left corticospinal tract

12. When α and γ motor neurons co-activate:

A. Inhibit Ib afferents
B. Clonus occurs
C. No contraction 18
D. Fewer Ib impulses are generated
E. More Ia impulses are generated than with α-activation alone

13. While walking on the beach, a patient suddenly steps on a sharp shell. Which sensory neuron/afferent fiber activates the flexor (withdrawal) reflex in the lower limb?

A. A-delta
B. Alpha motor neuron
C. C fiber
D. A-beta fiber
E. Ia afferent fiber

14. Given the table comparing static and dynamic responses in muscle spindles, which row is correct?

A. Static: Nuclear bag, Dynamic: Nuclear chain
B. Static: Nuclear chain & bag, Dynamic: Nuclear bag
C. Static: Nuclear chain, Dynamic: Nuclear bag & chain
D. Static: Nuclear bag & chain, Dynamic: Nuclear chain
E. Static: Nuclear chain, Dynamic: Nuclear bag

15. Which is NOT a sign of an upper motor neuron (UMN) lesion?

A. Hypertonia
B. Hyperreflexia
C. Clonus
D. Hemiparesis
E. Atrophy

16. Which of the following is NOT a pathological reflex?

A. Babinski sign
B. Clonus
C. Cremasteric reflex
D. Hoffman sign
E. Hyperreflexia

17. A patient has a lesion in the left pyramidal tract for one month. Which finding is NOT expected?

A. Hypertonia
B. Hyperreflexia
C. Paresis/paralysis
D. Muscular atrophy
E. Babinski sign

18. Which statement about muscle spindles is correct?

A. Bag = static; Chain = static
B. Bag = static & dynamic; Chain = static
C. Bag = dynamic; Chain = dynamic
D. Bag = static; Chain = static & dynamic
E. Bag = static & dynamic; Chain = dynamic

19. Which statement about the motor system is correct?

A. Lateral vestibulospinal tract facilitates extensor muscles.
B. Rubrospinal tract comes from the inferior colliculus.
C. Mammillary body comes from the dorsal tegmentum in the Midbrain.
D. Rubrospinal tract terminates in the ventral horn of the spinal cord to inhibit flexor muscles.
E. Medial Vestibulospinal tract comes from the lateral vestibular nucleus in the Caudal pons.

20. Which of the following is not a pathological reflex?

A. Cremasteric
B. Hoffman sign
C. Clonus
D. Babinski sign 31
E. Hyperreflexia (+4)

21. Which of the following is correct regarding the Achilles tendon reflex?

A. Receptor is muscle spindle
B. Effector is Hamstring
C. Afferent fiber is Golgi tendon organ
D. Reflex center is in L1-L2
E. Afferent and efferent both via the common peroneal nerve

22. Which of the following is not related to the motor system?

A. Skeletal muscle
B. Limbic lobe
C. Basal ganglia
D. Spinal cord
E. Cerebellum

23. Which statement about the motor system is correct?

A. Lateral vestibulospinal tract facilitates extensor muscles.
B. Rubrospinal tract comes from the inferior colliculus.
C. Mammillary body comes from the dorsal tegmentum in the Midbrain.
D. Rubrospinal tract terminates in the ventral horn of the spinal cord to inhibit flexor muscles.
E. Medial Vestibulospinal tract comes from the lateral vestibular nucleus in the Caudal pons.

24. According to the Neocortex, what is incorrect?

A. VPL terminates at layer 5.
B. Neocortex has 6 layers.
C. Corticospinal tract is an example of a projection fiber.
D. Association fibers connect gyri within the same hemisphere.
E. The neocortex makes up the majority of the human cerebral cortex.

25. Which of the following is correct regarding the Achilles tendon reflex?

A. Receptor is Golgi tendon organ
B. Effector is Hamstring
C. Afferent and efferent fibers are in the Tibial nerve
D. Reflex center is in L1-L2
E. Reflex center is in L4-L5

26. The reflexive head-turning toward a sudden sound is mediated predominantly by the:

A. Vestibulospinal tract
B. Reticulospinal tract
C. Tectospinal tract
D. Spinothalamic tract
E. Corticospinal tract

27. A lower motor neuron (LMN) lesion of the facial nerve results in:

A. Contralateral lower face paralysis
B. Ipsilateral lower face paralysis only
C. Entire ipsilateral facial paralysis
D. Contralateral upper face paralysis
E. Bilateral lower face weakness

Lecture 18 — Cerebellum (22 questions)

28. Which tract affects ataxia in this case?

A. Inferior spinocerebellar tract
B. Nucleus ambiguus
C. Olivocerebellar tract
D. Cuneocerebellar tract
E. Spinal trigeminal nucleus and tract

29. What is a sign of inferior cerebellar peduncle damage?

A. Ipsilateral ataxia
B. Ipsilateral hemiplegia
C. Contralateral paralysis
D. Contralateral pain/temp loss
E. Hypertonia ipsilateral

30. A 60-year-old woman uses melatonin for insomnia. Which organ excretes melatonin?

A. Liver
B. Kidney
C. Cerebellum
D. Sweat gland
E. Pineal gland

31. Which structure does NOT concern the stabilizing (vestibulocerebellar) circuit?

A. Pontine nuclei
B. Fastigial nucleus
C. Middle cerebellar peduncle
D. Superior cerebellar peduncle
E. Lateral dentate nucleus

32. Damage to which structure causes ataxia?

A. Superior cerebellar peduncle
B. Middle cerebellar peduncle
C. Inferior cerebellar peduncle
D. Dentate nucleus
E. Interposed nuclei

33. The output neuron of the cerebellar cortex is the:

A. Golgi cell
B. Granule cell
C. Basket cell
D. Stellate cell
E. Purkinje cell

34. Which of the following is not part of the cerebellum?

A. Basal ganglia
B. Skeletal muscle
C. Visual system
D. Brainstem
E. Cerebral cortex

35. The inability to perform rapid alternating movements combined with an intention tremor is known as:

A. Dysdiadochokinesia + intention tremor
B. Ataxia alone
C. Hemiballismus
D. Resting tremor
E. Chorea

36. A patient has a resting tremor that improves with movement and a dopamine deficiency. This suggests:

A. Parkinson’s disease
B. Essential tremor
C. Cerebellar tremor
D. Huntington’s disease
E. Wilson’s disease

37. A 72-year-old man has an inability to perform rapid alternating movements and an intention tremor, causing him to fall to the left. His condition is:

A. Dysdiadochokinesia and intention tremor
B. Ataxia alone
C. Hemiballismus
D. Resting tremor
E. Chorea

38. Which cerebellar deep nucleus is primarily for motor planning?

A. Fastigial
B. Globose
C. Emboliform
D. Lateral dentate
E. Medial dentate

39. Which of the following tracts to the cerebellum does not carry proprioception?

A. Dorsal spinocerebellar 26
B. Ventral spinocerebellar
C. Olivocerebellar
D. Tectocerebellar
E. Vestibulocerebellar

40. Which of the following tracts can inhibit the nucleus in the cerebellum 4 times per second?

A. Spinocerebellar tract
B. Olivocerebellar tract
C. Tectocerebellar tract
D. Vestibulocerebellar tract
E. Corticopontocerebellar tract

41. A damaged flocculonodular lobe of the cerebellum results in:

A. Intention tremor
B. Dysdiadochokinesia
C. Gait ataxia
D. Hypotonia
E. Resting tremor

42. Fibers from the dentate nucleus of the cerebellum travel to the _____ of the thalamus by passing through the _____.

A. VA / Superior cerebellar peduncle
B. VA / Inferior cerebellar peduncle
C. VL / Superior cerebellar peduncle
D. VL / Inferior cerebellar peduncle
E. VPL / Middle cerebellar peduncle

43. The stabilizing circuit (vestibulocerebellum) synapses with which nucleus in the cerebellum?

A. Dentate nucleus
B. Emboliform nucleus
C. Globose nucleus
D. Fastigial nucleus
E. Pontine nucleus

44. Which pair represents an excitatory input? 28

A. Purkinje cell - Golgi cell
B. Basket cell - Climbing fiber
C. Mossy fiber - Purkinje cell
D. Granular cell - Climbing fiber
E. Mossy fiber - Granular cell

45. The AICA (Anterior Inferior Cerebellar Artery) supplies all of the following EXCEPT: 30

A. Dorsolateral caudal pons
B. Inferior cerebellar peduncle
C. Middle cerebellar peduncle
D. Crus cerebri
E. Anterior surface of cerebellum

46. A patient has symptoms of dysdiadochokinesia and intention tremor. This is indicative of a lesion in the:

A. Cerebellum
B. Basal ganglia
C. Thalamus
D. Spinal cord
E. Peripheral nerve

47. A flocculonodular lobe lesion leads to:

A. Resting tremor 38
B. Dysdiadochokinesia
C. Gait ataxia
D. Dysarthria
E. Dysmetria

48. Astereognosis (inability to identify an object by touch) is caused by a lesion in the:

A. Primary motor cortex
B. Primary somatosensory cortex
C. Cerebellum
D. Basal ganglia
E. Occipital cortex

49. Climbing fibers to the cerebellum originate from the:

A. Red nucleus
B. Inferior olivary nucleus
C. Vestibular nuclei
D. Pontine nuclei
E. Reticular formation

Lecture 19 — Cerebral Cortex (31 questions)

50. A 73-year-old woman presents with right arm weakness and left gaze preference after recovery from a cerebrovascular accident. Sensory and pupillary light reflexes are normal. Her lesion is most likely found in which lobe?

A. Frontal
B. Parietal
C. Limbic
D. Occipital
E. Temporal

51. Which of the following pathways connects Wernicke's area to Broca's area?

A. Anterior commissure
B. Arcuate loops
C. Arcuate fasciculus
D. –
E. –

52. Which cortical cells provide axons to corpus callosum?

A. External granular
B. External pyramidal (pyramidal cells)
C. Internal granular
D. Internal pyramidal
E. Multiform

53. In stroke recovery, a 73-year-old woman has right arm weakness and left gaze preference. Which gyrus is most likely injured?

A. Left posterior middle frontal
B. Right posterior middle frontal
C. Left cuneus & lingual
D. Right cuneus & lingual
E. Left posterior middle frontal + cuneus & lingual

54. Which of the following is NOT a feature of Klüver-Bucy syndrome?

A. Lesions of the anterior thalamus
B. Hypersexuality
C. Hyperorality
D. Psychic blindness
E. Bilateral temporal lobe lesion

55. Olfactory information reaches the temporal lobe via:

A. Stria terminalis
B. Lateral olfactory stria
C. Orbital cortex
D. Anterior thalamic nucleus
E. Septal nuclei

56. A 38-year-old man has glioblastoma in the green-demarcated brain region. Which deficit occurs?

A. Korsakoff's syndrome
B. Klüver-Bucy syndrome
C. Broca's aphasia
D. Brown-Séquard syndrome
E. Horner's syndrome

57. A 63-year-old woman has spastic paralysis, right hemianesthesia, and a right facial droop, but no aphasia. The lesion is in the:

A. Right cortex
B. Base of pons (left)
C. Crus cerebri (left)
D. Pyramid of medulla (right)
E. Posterior limb & genu of left internal capsule

58. Which pathway most likely connects Wernicke's area to Broca's area in the left cerebral cortex?

A. Anterior commissure
B. Arcuate loops
C. Arcuate fasciculus
D. Corpus callosum
E. Uncinate fasciculus

59. A 73-year-old woman presents with right arm weakness and left gaze preference after a cerebrovascular accident. Sensory and pupillary reflexes are normal. Which lobe is most likely damaged?

A. Frontal
B. Parietal
C. Limbic
D. Occipital
E. Temporal

60. In the same patient from Q3 (right arm weakness, left gaze preference), which gyrus is injured?

A. Left posterior middle frontal
B. Right posterior middle frontal 12
C. Left cuneus & lingual
D. Right cuneus & lingual
E. Left posterior middle frontal + cuneus & lingual

61. A 71-year-old man with a left CVA presents with dysprosody, impaired singing, poor judgment, impulsiveness, lethargy, and indifference, but no aphasia. The lesion is in the:

A. Frontal lobe
B. Parietal lobe
C. Occipital lobe
D. Temporal lobe
E. Limbic lobe

62. Bilateral destruction of the amygdala causes:

A. Klüver-Bucy syndrome
B. Korsakoff's syndrome
C. Turner syndrome
D. Temporal lobe epilepsy
E. Horner's syndrome

63. Which cortical neurons provide axons to the corpus callosum?

A. External granular
B. External pyramidal
C. Internal granular
D. Internal pyramidal
E. Multiform layer

64. Ischemia of the paracentral lobule and cingulate gyrus results from occlusion of the:

A. Frontopolar artery
B. Callosomarginal artery
C. Precentral artery
D. Paracentral artery
E. Calcarine artery

65. A 71-year-old man presents with contralateral hemineglect, anosognosia, and visuospatial disability, but no aphasia. Which lobe is damaged?

A. Frontal
B. Parietal
C. Occipital
D. Temporal
E. Limbic

66. The habenulointerpeduncular tract projects to the:

A. Midbrain
B. Thalamus
C. Limbic lobe
D. Insular cortex
E. Neurohypophysis

67. A patient has abnormal arm and leg movements but remains fully conscious. This is a:

A. Simple partial seizure
B. Complex partial seizure
C. Absence seizure
D. Temporal lobe epilepsy
E. Generalized tonic-clonic seizure

68. Damage to the paracentral lobule and cingulate gyrus is caused by occlusion of which artery?

A. Callosomarginal artery
B. Paracentral artery
C. Middle cerebral artery
D. Anterior cerebral artery
E. Posterior cerebral artery

69. A patient has finger agnosia and dyscalculia. The lesion is in the:

A. Left angular gyrus
B. Right angular gyrus
C. Left precuneus
D. Right precuneus
E. Left supramarginal gyrus

70. A patient presents with hypersexuality, hyperorality, and placidity. The lesion is in the:

A. Dorsomedial thalamus
B. Anterior temporal lobe
C. Posterior hypothalamus
D. Cingulate gyrus
E. Septal nuclei

71. An MRI shows a lesion in the right pars opercularis. The patient will likely have what disorder?

A. Motor apraxia
B. Motor aphasia
C. Dysarthria
D. Right hemiplegia
E. Dysprosody

72. A lesion in the frontal, parietal, and temporal lobes of the non-dominant hemisphere would most likely cause:

A. Alexia
B. Broca's Aphasia
C. Wernicke's Aphasia
D. Dysprosody
E. Prosopagnosia

73. A patient has right-sided neglect but no aphasia. The lesion is likely in the:

A. Frontal lobe
B. Temporal lobe
C. Limbic lobe
D. Occipital lobe
E. Parietal lobe

74. A patient has abnormal movement in their right arm and does not lose consciousness. What is the best description of the patient's condition?

A. Simple partial seizure
B. Complex partial seizure
C. Tonic-clonic seizure
D. Absence seizure
E. Temporal lobe epilepsy

75. A patient has abnormal movement in their right arm but loses consciousness. What is the best description of the patient's condition?

A. Simple partial seizure
B. Complex partial seizure
C. Tonic-clonic seizure
D. Absence seizure
E. Temporal lobe epilepsy ■ MED33 (Set 2)

76. A lesion in the left cerebral cortex affects the corticobulbar fibers. What is true according to the lesion?

A. Right ptosis
B. Right inability to close eyelids
C. Right weakness in closing mouth tightly
D. Left inability to raise shoulder
E. Right tongue deviation

77. A patient exhibits hyperorality and hypersexuality. This is characteristic of:

A. Klüver-Bucy syndrome
B. Korsakoff's syndrome
C. Gerstmann syndrome
D. Frontal lobe syndrome
E. Wernicke's aphasia

78. The fiber tract connecting Broca’s and Wernicke’s areas is the:

A. Corpus callosum
B. Internal capsule
C. Arcuate fasciculus
D. Superior longitudinal fasciculus
E. Uncinate fasciculus

79. A patient understands language but cannot speak fluently. The lesion is at:

A. Wernicke's area
B. Angular gyrus
C. Broca's area
D. Arcuate fasciculus
E. Precentral gyrus

80. Alexia without agraphia (pure word blindness) is caused by a lesion in the:

A. Broca's area
B. Angular gyrus
C. Splenium of the corpus callosum
D. Wernicke's area
E. Precentral gyrus

Lecture 20 — Basal Ganglia (11 questions)

81. The globus pallidus projects to the thalamus via:

A. Ansa lenticularis
B. Ansa peduncularis
C. Fasciculus retroflexus
D. Stria medullaris
E. Stria terminalis

82. Which fibers are excitatory in the basal ganglia pathway?

A. GPe → STN
B. GPi → VL thalamus 5
C. STN → GPi
D. Striatum → GPi
E. Striatum → GPe

83. A 25-year-old woman collapses with tonic stiffening for 30 seconds, followed by jerking for 2 minutes. What is the diagnosis?

A. Complex partial seizure
B. Simple partial seizure
C. Tonic-clonic seizure
D. Absence seizure
E. Parkinson's disease

84. A woman suddenly collapses with tonic stiffening followed by clonic jerks. The diagnosis is:

A. Complex partial seizure
B. Simple partial seizure
C. Tonic-clonic seizure
D. Absence seizure
E. Parkinson's disease

85. Which basal ganglia structure is in the midbrain?

A. Caudate
B. Putamen
C. Subthalamic nucleus
D. Globus pallidus externus
E. Substantia nigra (pars reticulata)

86. A 65-year-old man has a resting tremor, bradykinesia, shuffling gait, and decreased dopamine on a PET scan. The diagnosis is:

A. Hemiballismus
B. Tardive dyskinesia
C. Sydenham’s chorea
D. Parkinson’s disease
E. Huntington’s disease

87. What is the most common cause of hemiballismus–hemichorea?

A. Stroke
B. Seizure
C. Brain tumor
D. Multiple sclerosis
E. Traumatic brain injury

88. A patient with uncontrolled movements has atrophy of the basal ganglia. Her mother had similar symptoms. What is her condition?

A. Bradykinesia
B. Parkinson Disease
C. Hypokinetic disorder
D. Huntington's Disease
E. Sydenham's chorea

89. A patient has Parkinson's disease. What is the effect on the basal ganglia pathways?

A. Overactivity of the direct pathway, underactivity of the indirect pathway
B. Underactivity of the direct pathway, overactivity of the indirect pathway
C. Increased excitatory effect of the direct pathway
D. Increased inhibition of the indirect pathway
E. No change in either pathway

90. Which of the following indicates Parkinson's disease more specifically than other movement disorders?

A. Unilateral resting tremor
B. MRI findings
C. Chorea
D. Intention tremor
E. Bilateral onset of symptoms

91. Which supraspinal site is the key initiator of descending analgesia recruited by opioids?

A. Solitary nucleus
B. Substantia nigra
C. Periaqueductal gray (PAG)
D. Ventral posterior thalamus
E. Primary motor cortex

Lecture 21 — Hypothalamus & Epithalamus (10 questions)

92. A 30-year-old man had mammillary body damage. Which symptom is found?

A. Hyperorality
B. Hypersexuality
C. Facial blunting
D. Loss of recent memory
E. Loss of long-term memory

93. A 30-year-old man with mammillary body damage would present with:

A. Hyperorality
B. Hypersexuality
C. Facial blunting
D. Loss of recent memory
E. Loss of long-term memory

94. Which hypothalamic nucleus lies in the tuberal region?

A. Arcuate
B. Supraoptic
C. Mammillary
D. Paraventricular
E. Medial preoptic

95. A 35-year-old man presents with constant hunger, weight loss, and poor concentration. Which hypothalamic nucleus is damaged?

A. Arcuate
B. Supraoptic
C. Mammillary
D. Paraventricular
E. Ventromedial nucleus

96. What increases melatonin synthesis?

A. Inhibited suprachiasmatic nucleus
B. Inhibited paraventricular nucleus
C. Stimulated superior cervical ganglion
D. Parasympathetic control 21
E. Stimulated suprachiasmatic nucleus

97. Which nucleus lies in the supraoptic region of the hypothalamus?

A. Paraventricular
B. Centromedian
C. Posterior
D. Mammillary
E. Dorsolateral

98. Jet lag is due to dysregulation of the:

A. Pineal gland
B. Suprachiasmatic nucleus
C. Paraventricular nucleus
D. Supraoptic nucleus
E. Locus coeruleus

99. Which statement about the functions of hypothalamic nuclei and their tracts is correct?

A. Suprachiasmatic nuclei can send signals back to the retina via the optic tract.
B. The Mammillothalamic tract connects the dorsomedial nuclei and the mammillary body.
C. The posterior hypothalamic nucleus sends sympathetic signals to the spinal cord (T1-L2) via the hypothalamospinal tract.
D. The anterior hypothalamic nucleus cannot send parasympathetic signals to the Edinger-Westphal nucleus. 29
E. The preoptic area is involved in thirst.

100. What increases the synthesis of melatonin?

A. Inhibited suprachiasmatic nucleus
B. Inhibited paraventricular nucleus
C. Stimulation of optic tract
D. Inhibited superior cervical ganglion
E. Its control by parasympathetic system

101. The heat gain center in the hypothalamus is located in the:

A. Anterior nucleus
B. Posterior nucleus
C. Arcuate nucleus
D. Suprachiasmatic nucleus
E. Lateral hypothalamus

Lecture 22 — Thalamus (14 questions)

102. Which thalamic nucleus receives input from contralateral lateral spinothalamic tract?

A. Centromedian nucleus
B. Mediodorsal nucleus
C. Ventral anterior nucleus
D. Ventral lateral nucleus
E. Ventral posterolateral nucleus

103. Which sensation bypasses the thalamus before reaching the cortex?

A. Vision
B. Hearing
C. Fine touch
D. Pain
E. Olfaction

104. Which sensation does not synapse in the thalamus before reaching the cortex?

A. Vision
B. Hearing
C. Fine touch
D. Pain
E. Olfaction

105. Which thalamic nucleus connects to the solitariothalamic tract? 6

A. Centromedian nucleus
B. Mediodorsal nucleus
C. Ventral anterior nucleus
D. Ventral lateral nucleus
E. Ventral posteromedial nucleus

106. Which thalamic nuclei influence activity of the motor cortex?

A. Dorsomedial & lateral dorsal nuclei
B. Ventral anterior & ventral lateral nuclei
C. Lateral posterior & ventral posteromedial nuclei
D. Intralaminar & midline nuclei
E. Lateral geniculate & reticular nuclei

107. Which thalamic nucleus receives input from the contralateral lateral spinothalamic tract?

A. Centromedian nucleus
B. Mediodorsal nucleus 7
C. Ventral anterior nucleus
D. Ventral lateral nucleus
E. Ventral posterolateral nucleus

108. Which sensation bypasses the thalamus before reaching the cortex?

A. Vision
B. Hearing
C. Taste
D. Touch
E. Smell

109. Which thalamic nuclei influence the motor cortex?

A. Dorsomedial & lateral dorsal
B. Ventral anterior & ventral lateral
C. Lateral posterior & ventral posteromedial
D. Intralaminar & midline
E. Lateral geniculate & reticular

110. The thalamic nucleus receiving input from the lateral spinothalamic tract is the:

A. Centromedian
B. Mediodorsal
C. Ventral anterior
D. Ventral lateral
E. Ventral posterolateral

111. The thalamic nucleus connected to the solitariothalamic tract is the:

A. Centromedian
B. Mediodorsal
C. Ventral anterior
D. Ventral lateral
E. Ventral posteromedial

112. Which of the following is correct regarding the touch sensation pathway?

A. The third-order neuron synapses in the thalamus.
B. The second-order neuron synapses in the cuneate nucleus of the brainstem.
C. The final neuron projects to the primary somatosensory area.
D. The first-order neuron is in the ventral horn. ■ 36 MCQs (Finalized, Clean)
E. The fourth-order neuron is located in the dorsal root ganglion.

113. The gustatory pathway includes all of the following EXCEPT:

A. Solitary nucleus (rostral part)
B. VPM nucleus of the thalamus
C. Geniculate/petrosal/nodose ganglia
D. Insula and frontal operculum (gustatory cortex)
E. Trapezoid body

114. Within the Papez circuit, damage to which thalamic nucleus impairs recent memory?

A. VPM
B. Dorsomedial
C. Anterior nucleus
D. Pulvinar
E. Lateral geniculate

115. A lesion of the anterior thalamic nucleus causes:

A. Astereognosis
B. Anterograde amnesia
C. Dysgeusia
D. Hemineglect
E. Dysarthria

Lecture 23 — Limbic System & Olfaction (12 questions)

116. What is the relationship of the hippocampus to the inferior horn of the lateral ventricle?

A. In floor
B. In roof
C. Rostral to ventricle
D. In front
E. No relation

117. A 39-year-old woman has an eating disorder, thirst, and fluctuating temperature. Which structure is dysfunctional?

A. Amygdala
B. Hippocampus
C. Hypothalamus
D. Septal nucleus
E. Habenula

118. Which is NOT a major bidirectional hypothalamic pathway?

A. Fornix from hippocampus
B. Medial forebrain bundle → septal area
C. Dorsal longitudinal fasciculus → PAG
D. Retinohypothalamic tract → SCN
E. Medial forebrain bundle & dorsal longitudinal fasciculus → brainstem & spinal cord

119. A 39-year-old woman has an eating disorder, thirst, and temperature dysregulation. This indicates dysfunction of the:

A. Amygdala
B. Hippocampus 8
C. Hypothalamus
D. Septal nucleus
E. Habenula nucleus

120. The conversion of short-term to long-term memory occurs in the:

A. Thalamus
B. Basal ganglia
C. Hypothalamus
D. Amygdala
E. Hippocampus

121. Sara feels fear and has an increased heart rate after hearing a sound at night. Which brain part is most active?

A. Thalamus
B. Basal ganglia
C. Hypothalamus
D. Amygdala
E. Hippocampus

122. A 65-year-old alcoholic presents with severe memory impairment. Atrophy is expected in which structure?

A. Cingulate gyrus
B. Parahippocampal gyrus
C. Hippocampus
D. Amygdala
E. Mammillary body

123. A patient has an eating disorder, severe thirst, and a wildly varying body temperature. Which structure is most likely dysfunctional?

A. Putamen
B. Habenula
C. Amygdala
D. Hippocampus
E. Hypothalamus

124. An alcoholic patient has a memory deficit. Which structure is most likely damaged?

A. Amygdaloid Complex
B. Mammillary Body
C. Hippocampus
D. Septal Nuclei
E. Locus coeruleus

125. Which of the following is correct about the fornix?

A. Precommissural fornix connects to the anterior nucleus of the thalamus.
B. Postcommissural fornix connects to the mammillary body.
C. The fornix originates from the amygdala.
D. The fornix terminates in the cingulate gyrus.
E. The fornix connects directly to the amygdala without passing through any commissure.

126. An alcoholic with a memory deficit most likely has damage to the:

A. Mammillary Body
B. Amygdala
C. Hippocampus
D. Cingulate gyrus
E. Nucleus accumbens

127. Which stem cell type regenerates sensory receptor cells in the tongue and olfactory epithelium?

A. Satellite cells
B. Schwann cells
C. Basal cells
D. Stem keratinocytes
E. Merkel cells

Lecture 24 — Autonomic Nervous System I (Structures) (10 questions)

128. Throughout the autonomic nervous system, preganglionic fibers release neurotransmitter that binds to which receptor? 3

A. N 2
B. M 3
C. α 1
D. α 2
E. β 2

129. Which is NOT a feature of Horner’s syndrome?

A. Ptosis
B. Anhydrosis
C. Miosis
D. Enophthalmos
E. No saliva secretion

130. A lesion in which location can present with Horner’s syndrome?

A. Ciliary ganglion
B. Stellate ganglion
C. Superior cervical ganglion
D. Middle cervical ganglion
E. Sympathetic chain T1–T4 ■ MED34

131. Which of the following nerves carries parasympathetic fibers?

A. Lumbar splanchnic
B. Greater splanchnic
C. Lesser splanchnic
D. Pelvic splanchnic
E. Least splanchnic

132. In the autonomic nervous system, preganglionic fibers release a neurotransmitter that binds to:

A. N2 receptors
B. M3 receptors
C. α1 receptors
D. α2 receptors
E. β2 receptors

133. Which of the following does NOT form the cardiac plexus?

A. Superior cardiac branch
B. Middle cardiac branch
C. Inferior cardiac branch
D. Postganglionic fibers T1–T4
E. Greater splanchnic nerve

134. Which sympathetic/parasympathetic action is correct?

A. Pupils: Symp constrict, Para dilate
B. Heart: Symp ↓HR, Para ↑HR
C. Sweat glands: Symp ↓, Para ↑
D. Bladder: Symp relaxes, Para contracts
E. Male organ: Symp erection, Para ejaculation

135. Which pairing of preganglionic neurons and their fibers is incorrect?

A. Edinger–Westphal → Oculomotor nerve
B. Superior salivatory → Greater petrosal nerve
C. Inferior salivatory → Lesser petrosal nerve
D. Dorsal motor nucleus of vagus → CN X
E. T1–T4 → Superior cardiac branch

136. The primary source of circulating epinephrine is:

A. Postganglionic sympathetic neurons
B. Preganglionic sympathetic neurons
C. Adrenal medullary chromaffin cells
D. Parasympathetic postganglionic neurons
E. Preganglionic parasympathetic neurons

137. Which of the following is a parasympathetic fiber?

A. Least splanchnic nerve
B. Lumbar splanchnic nerve
C. Sacral splanchnic nerve
D. Pelvic splanchnic nerve
E. Greater splanchnic nerve

Lecture 25 — Autonomic Nervous System II (Functions) (4 questions)

138. A patient has wild mushroom poisoning (muscarinic). Expected findings are:

A. Increased salivation, bradycardia, relaxed urinary sphincter
B. Decreased tears, tachycardia, contracted urinary sphincter
C. Increased tears, bradycardia, relaxed urinary sphincter 20
D. Increased salivation, tachycardia, contracted urinary sphincter
E. Increased tears, bradycardia, contracted urinary sphincter ■ MED33 SA2

139. Which enzyme converts norepinephrine to epinephrine in the adrenal medulla?

A. Choline acetyltransferase
B. Acetylcholinesterase
C. Phenylethanolamine N-methyltransferase (PNMT)
D. Epinephrine synthase
E. AA-NAT

140. Norepinephrine for arousal and alertness is synthesized mainly in the:

A. Raphe nuclei
B. Substantia nigra
C. Locus coeruleus
D. Basal forebrain
E. Hypothalamus

141. The enzyme that converts norepinephrine (NE) to epinephrine (Epi) is:

A. COMT
B. MAO 40
C. PNMT
D. Tyrosine hydroxylase
E. Dopamine β-hydroxylase

Lecture 26 — EEG & Sleep Physiology (11 questions)

142. What are the characteristics of an EEG α-rhythm?

A. Deep sleep
B. 20–30 Hz 4
C. Disappears with eye opening
D. Replaced by slower waves in REM
E. Appears in normal waking state

143. A 10-year-old boy has a generalized seizure with momentary loss of consciousness. His EEG shows 2.5–3 Hz generalized spike-wave activity. What is the diagnosis?

A. Clonic
B. Absence
C. Tonic-clonic
D. Simple partial
E. Complex partial

144. A 35-year-old man has obstructive sleep apnea. Which EEG pattern occurs during REM sleep?

A. High amplitude & frequency
B. Low amplitude & frequency 10
C. High amplitude, low frequency
D. Low amplitude, high frequency
E. K complexes & spindles

145. A 35-year-old man transitions from non-REM sleep to wakefulness. Which neurotransmitter change occurs?

A. ↓NE, ↑5-HT, ↑ACh, ↓Histamine, ↓GABA
B. ↓NE, ↑5-HT, ↑ACh, ↓Histamine, ↑GABA
C. ↓NE, ↓5-HT, ↑ACh, ↑Histamine, ↑GABA
D. ↑NE, ↑5-HT, ↑ACh, ↑Histamine, ↓GABA
E. ↑NE, ↓5-HT, ↓ACh, ↑Histamine, ↓GABA

146. The EEG α-rhythm is characterized by:

A. Association with deep sleep
B. 100–120 Hz
C. Disappearance when eyes open
D. Replacement by slower waves in REM
E. Appearance in normal waking state

147. A 10-year-old boy has brief losses of consciousness. His EEG shows 2.5–3 Hz spike-wave discharges. What is the diagnosis? 16

A. Clonic seizures
B. Absence seizures
C. Tonic-clonic seizures
D. Simple partial seizures
E. Complex partial seizures

148. The EEG in REM sleep shows:

A. High amplitude, high frequency
B. Low amplitude, low frequency
C. High amplitude, low frequency
D. Low amplitude, high frequency
E. K complexes & spindles

149. What are the changes in neurotransmitters when transitioning from NREM sleep to waking?

A. Locus Coeruleus ↓NE & SE, Raphe nuclei ↑ACh
B. Locus Coeruleus ↑NE & SE, Raphe nuclei ↓ACh
C. Locus Coeruleus ↓Histamine, Raphe nuclei ↑GABA
D. Locus Coeruleus ↓NE, Raphe nuclei ↓5-HT
E. Locus Coeruleus ↑Histamine only, no other changes

150. Which brainwave pattern occurs in deep sleep (Stage N3)?

A. Gamma
B. Beta
C. Alpha
D. Theta
E. Delta

151. The EEG waveform in Stage 4 (N3) sleep is characterized by:

A. K complex & spindles
B. High amplitude, low frequency
C. High amplitude, high frequency
D. Low amplitude, high frequency
E. Low amplitude, low frequency

152. Which statement about sleep stages is incorrect?

A. Stage 1: drowsy and eyes close
B. Stage 2: sleep spindle
C. Stage 3: delta and theta wave
D. Stage 4: slow-wave-sleep
E. REM sleep: night terror ■ MED32

Lecture 27 — Reflex Mechanism (6 questions)

153. Which receptor mediates corneal reflex sensation?

A. Merkel disks
B. Ruffini endings
C. Free nerve endings
D. Pacinian corpuscles
E. Meissner corpuscles ■ MED35

154. The sensory receptor for the corneal reflex is the:

A. Merkel disks
B. Ruffini endings
C. Free nerve endings
D. Pacinian corpuscles
E. Meissner corpuscles

155. The cause of the rebound phenomenon (a cerebellar sign) is:

A. No stretch reflex 14
B. Postural disturbance
C. Muscle spasm
D. Areflexia
E. Loss of muscle power

156. The Glasgow Coma Scale (GCS) consists of assessing:

A. Motor, verbal responses, eye opening
B. Reflexes, orientation, motor responses
C. Pupillary response, verbal, pain response
D. Temperature, pulse, blood pressure
E. Cranial nerve function, strength, coordination

157. In the pupillary light reflex, sensory input is via CN ___ and motor output is via CN ___.

A. II, III
B. II, IV
C. III, V
D. V, VII
E. III, VI

158. The gag reflex is mediated by which brainstem level?

A. Midbrain
B. Pons
C. Medulla oblongata
D. Diencephalon
E. Cervical cord

Lecture 28 — Blood Supply of Brain & Meninges (31 questions)

159. You are caring for a patient who has suffered a stroke and has issues comprehending spoken language (Aphasia). Branches of which artery are compromised?

A. ACA
B. MCA
C. PCA
D. Striate artery
E. Anterior choroidal artery

160. Cerebrospinal fluid communicates with the subarachnoid space via which structure?

A. 4th ventricle
B. 3rd ventricle
C. Subarachnoid granulations
D. Choroid plexus
E. Cerebral aqueduct

161. A 55-year-old male, post-stroke, has a shaded lesion in the lower pons (autopsy diagram). The embolus is in which artery?

A. ACA
B. MCA
C. PICA
D. AICA
E. Basilar artery

162. Which artery is most likely involved in this lesion?

A. Basilar artery
B. Superior cerebellar artery
C. Posterior cerebral artery
D. Posterior inferior cerebellar artery
E. Anterior inferior cerebellar artery

163. Regarding cerebral cortex blood supply, which is true?

A. MCA: contralateral arm, leg, speech areas
B. ACA: contralateral leg and micturition
C. MCA: ipsilateral arm, face, vision
D. PCA: ipsilateral vision
E. ACA: contralateral leg, auditory, speech

164. Superior sagittal sinus is located between:

A. Inner table of skull & endosteal dura
B. Endosteal dura & meningeal dura
C. Meningeal dura & arachnoid
D. Arachnoid & pia
E. Pia mater & arachnoid mater

165. Which statement about strokes is false?

A. Stroke = occlusion/rupture
B. TIA lasts 2–3 weeks
C. Ischemic stroke = rupture of vessel
D. Hemorrhagic stroke = fat/plaque
E. AVMs = clot forming in vessel

166. A 65-year-old male was admitted after a stroke and later died. The autopsy cross-section of his pons shows a dark lesion. The embolus most likely blocked which artery?

A. ACA
B. MCA
C. PICA
D. Posterior communicating artery
E. Basilar artery

167. Which of the following statements about stroke is true?

A. TIA often lasts 2–3 weeks
B. Ischemic stroke = rupture of brain blood vessels
C. Hemorrhagic stroke = due to plaques/fat lining
D. AVMs = local cerebral clots
E. Stroke = neurological injury from occlusion or rupture of cerebral vessels

168. An occlusion of the right ACA (anterior cerebral artery) will cause deficits in:

A. Motor & sensory in right lower limbs
B. Motor & sensory in left lower limbs
C. Motor & sensory in right upper limbs
D. Motor & sensory in left upper limbs
E. Motor & sensory in all limbs

169. A 46-year-old man presents with apathy, severe right leg weakness greater than arm/face weakness, and urinary incontinence. Which artery is mainly occluded?

A. Lateral striate artery
B. Right MCA
C. Left MCA
D. Right ACA
E. Left ACA

170. Regarding cortical blood supply, which statement is true?

A. PCA → ipsilateral deafness
B. MCA → ipsilateral arm, face, vision
C. ACA → contralateral leg & micturition
D. ACA → contralateral leg, auditory, speech
E. MCA → contralateral face, arm, leg, speech areas

171. Which artery is closely related to the primary visual area?

A. Calcarine artery
B. Frontopolar artery
C. Paracentral artery
D. Prerolandic artery
E. Callosomarginal artery

172. An MRI shows a homogeneous pontine cistern mass. Which vessel is most affected?

A. Basilar artery
B. Anterior spinal artery
C. Middle cerebral artery
D. Posterior communicating artery
E. Posterior inferior cerebellar artery

173. Aphasia due to artery occlusion is most often from the:

A. ACA
B. MCA
C. PCA
D. AICA
E. Anterior choroidal artery

174. Right ACA occlusion causes a deficit in:

A. Right leg motor/sensory
B. Left leg motor/sensory
C. Right arm motor/sensory
D. Left arm motor/sensory
E. All limbs

175. A 58-year-old male presents with right leg weakness greater than arm/face weakness, plus apathy. Which artery is occluded?

A. Lateral striate
B. Right MCA
C. Left MCA
D. Right ACA
E. Left ACA

176. A 55-year-old man has a lesion in the pons found on autopsy. The embolus was likely in the:

A. ACA
B. MCA
C. PICA
D. Posterior communicating artery
E. Basilar artery

177. The artery most related to the primary visual area is the:

A. Calcarine artery
B. Frontopolar artery
C. Paracentral artery
D. Prerolandic artery
E. Callosomarginal artery

178. A 71-year-old man has a large left MCA stroke. Which artery is most likely affected?

A. ACA
B. MCA
C. PCA
D. ICA
E. Vertebral

179. Which statement is true?

A. Stroke = occlusion or rupture of cerebral vessels
B. TIA = 2–3 weeks
C. Ischemic stroke = rupture of vessels
D. Hemorrhagic stroke = due to plaques
E. AVMs = clots form locally

180. A pituitary tumor typically causes which visual defect?

A. Optic nerve lesion 22
B. Optic chiasm lesion
C. Optic tract lesion
D. Parietal upper optic radiation lesion
E. Temporal lower optic radiation lesion

181. What is the first color typically lost in color blindness?

A. Red
B. Green
C. Purple
D. Yellow
E. Blue

182. A patient has contralateral sensory loss and ipsilateral CN III palsy. This is caused by occlusion of the:

A. ACA
B. MCA
C. PCA
D. Anterior choroidal artery
E. Basilar artery

183. Which statement about venous drainage is correct?

A. Superior sagittal sinus → great cerebral vein
B. Superior cerebral veins → superior sagittal sinus
C. Inferior sagittal sinus → cavernous sinus
D. Great cerebral vein → superior sagittal sinus
E. Cavernous sinus → superior sagittal sinus directly

184. An ischemic stroke in the pontine cistern involves which artery?

A. Basilar artery
B. Calcarine artery
C. ACA
D. PCA
E. MCA

185. A patient has an ischemic stroke in the pontine cistern. Which artery is likely to be occluded?

A. Basilar artery
B. Calcarine artery
C. Anterior cerebral artery
D. Posterior cerebral artery
E. Middle cerebral artery

186. Which of the following is NOT a criterion used in the ABCD2 score for stroke risk after a TIA?

A. Hyperlipidemia
B. Diabetes
C. Age
D. Clinical feature
E. Blood pressure (Hypertension)

187. An embolus from the internal carotid artery is most likely to lodge in which branch?

A. Middle cerebral artery (MCA)
B. Posterior communicating artery (PcomA)
C. Anterior cerebral artery (ACA)
D. Anterior communicating artery (AcomA)
E. Posterior cerebral artery (PCA)

188. A patient has an embolus from the internal carotid artery. In which branch is the embolus most likely to lodge?

A. MCA
B. PCOM
C. ACOM
D. ACA
E. VA

189. The arterial supply of the primary visual cortex is the:

A. Middle cerebral artery
B. Anterior cerebral artery
C. Calcarine artery (PCA branch)
D. Superior cerebellar artery
E. Anterior choroidal artery

Lecture 29 — Clinical Correlations (1 questions)

190. A 37-year-old woman presents with fatigue, difficulty speaking, ptosis, and diplopia that worsens throughout the day. Sensory exam is normal. What is the diagnosis?

A. Multiple sclerosis
B. Guillain-Barre syndrome
C. Lambert-Eaton Syndrome
D. Tetanus
E. Myasthenia Gravis

Mixed / Other Topics (35 questions)

191. A 50-year-old man has ptosis, anhidrosis, and miosis due to a lung apex tumor. Which muscle is linked to the drooping eyelid?

A. Ciliary muscle
B. Pupillary sphincter
C. Pupillary dilator
D. Superior tarsal muscle
E. Arrector pili

192. A 50-year-old man with a lung apex tumor presents with miosis. Which muscle correlates with this finding?

A. Ciliary muscle
B. Arrector pili
C. Pupillary dilator
D. Superior tarsal
E. Hamstring

193. A 40-year-old man has presbyopia (difficulty reading). Which statement about near focusing is correct?

A. Lens becomes flatter
B. Controlled by sympathetic NS
C. M3 receptors contract ciliary muscle
D. $\alpha 2$ receptors contract sphincter muscle
E. Ciliary contraction increases suspensory tension

194. A 30-year-old woman has a UTI. Which receptor contracts the detrusor muscle?

A. β 2
B. β 3
C. N
D. M 2
E. M 3

195. Which of the following is a sympathetic effect?

A. Pupil constriction
B. Increased heart rate
C. Increased small intestine contraction
D. Limb vessel dilation
E. Increased urination

196. Which receptor mediates detrusor muscle contraction for urination?

A. β2
B. β3
C. N
D. M2
E. M3

197. What is the best diagnostic test for Huntington's disease?

A. CT scan
B. MRI
C. Genetic testing
D. Blood glucose
E. EMG (electromyography)

198. Huntington's disease involves the loss of which neurotransmitter?

A. GABA
B. Serotonin
C. Dopamine
D. Substance P
E. Acetylcholine

199. Which sympathetic receptor is used to contract the internal urinary sphincter?

A. Alpha-1
B. Beta-3
C. Beta-2
D. NM
E. M3

200. What is the maximum GCS score?

A. 10
B. 12
C. 15
D. 18
E. 20

201. The nerve roots for the flexors of the arm (e.g., biceps) are:

A. C5–C6
B. L1–L2
C. L5–S1
D. S2–S3
E. T1–T2

202. The brain waves in an awake, alert state with eyes open are:

A. Delta
B. Beta
C. Alpha
D. Gamma
E. Theta

203. A left corticobulbar lesion would result in which finding?

A. Right ptosis
B. Inability to close right eyelid
C. Weakness closing mouth tightly (right)
D. Inability to raise left shoulder
E. Tongue deviates right ■ MED31

204. A patient has difficulty breathing and weakness in facial expression. Which part of the internal capsule might be affected?

A. Genu
B. Posterior limb
C. Anterior limb
D. Sublenticular part
E. Retrolenticular part

205. What type of fiber provides input to the neostriatum?

A. Nigrostriatal, inhibitory via GABA
B. Corticostriatal, excitatory via Glutamate
C. Nigrostriatal, excitatory via Glutamate
D. Thalamostriatal, inhibitory via GABA
E. Pallidostriatal, inhibitory via GABA

206. A patient presents with vertical diplopia and has downward gaze in the left eyeball when she adducts her eyes. What is the cranial nerve lesion?

A. Left CN IV
B. Left CN III
C. Right CN II
D. Left CN VI
E. Right CN VI

207. What is the primary receptor for detrusor muscle contraction during urination?

A. Alpha-1
B. Alpha-2
C. Beta-1
D. Beta-2
E. M3

208. A patient comes with vertical diplopia and has difficulty looking down with the left eye, especially when the eye is adducted. What is the cranial nerve lesion?

A. Left CN IV
B. Left CN III
C. Right CN II
D. Left CN VI
E. Right CN VI

209. The ganglia of the first-order taste neurons are located in which of the following structures?

A. Geniculate, Petrosal, Nodose ganglia
B. Superior, Ciliary, Otic ganglia
C. Stellate, Geniculate, Nodose ganglia
D. Trigeminal, Geniculate, Nodose ganglia
E. Geniculate, Spiral, Pterygopalatine ganglia

210. An unpleasant or distorted taste sensation is called:

A. Dysphagia
B. Dysphonia
C. Dysgeusia
D. Dysarthria
E. Dysmetria

211. When stereocilia of a hair cell bend toward the kinocilium:

A. Hyperpolarization occurs
B. K+ channels close
C. Depolarization occurs
D. Ca2+ efflux increases
E. Action potential frequency decreases

212. In the Gate-control theory of pain, enkephalins act by:

A. Blocking Na+ channels presynaptically
B. Opening presynaptic Ca2+ channels
C. Inhibiting Ca2+ influx and opening K+ channels
D. Stimulating NMDA receptors 35
E. Enhancing substance P release

213. Which neurotransmitter-transporter pair is CORRECT?

A. Glutamate — VGAT / EAAT
B. GABA — VGLUT / GAT
C. Serotonin — VMAT / SERT
D. Dopamine — SERT / EAAT
E. Glycine — DAT / GAT

214. Long-Term Potentiation (LTP) requires all of the following EXCEPT:

A. NMDA receptor Ca2+ influx
B. Insertion of AMPA receptors into the membrane
C. Mg2+ block removal from the NMDA receptor
D. Retrograde NO messenger
E. Activation of GABA-A receptors

215. The mechanism of action for diazepam (a benzodiazepine) is:

A. Blocks GABA reuptake
B. Increases the duration of Cl– channel opening
C. Increases the frequency of Cl– channel opening at the GABA-A receptor
D. Blocks NMDA receptor Ca2+ influx
E. Enhances glycine binding

216. Which of the following is NOT a treatment for myasthenia gravis?

A. Anticholinesterase drugs
B. Prednisone
C. Plasmapheresis
D. Thymectomy
E. Dopamine agonists

217. The neurotransmitter primarily linked to mood regulation is:

A. Dopamine
B. GABA
C. Serotonin
D. Glutamate
E. Glycine

218. A lesion of the left Meyer's loop (temporal optic radiation) results in: 36

A. Left homonymous hemianopia
B. Right superior quadrantanopia
C. Right inferior quadrantanopia
D. Bitemporal hemianopia
E. Left superior quadrantanopia

219. A midline lesion of the optic chiasm (e.g., from a pituitary tumor) causes:

A. Right homonymous hemianopia
B. Left homonymous hemianopia
C. Right superior quadrantanopia
D. Bitemporal hemianopia
E. Binocular blindness

220. In the semicircular canal system, during which condition does afferent firing return to baseline?

A. Angular acceleration
B. Angular deceleration
C. Constant-velocity rotation
D. Onset of head rotation
E. Head oscillation

221. The primary receptor for cold sensation is:

A. TRPV1
B. TRPA1
C. TRPM8
D. ASIC
E. ENaC

222. Which nerve does NOT synapse in the spinal trigeminal nucleus?

A. CN V
B. CN VII
C. CN IX
D. CN X
E. CN XI

223. The mechanism of action of fluoxetine is:

A. Blocks VMAT
B. Inhibits COMT
C. Inhibits MAO-A
D. Inhibits SERT
E. Enhances GABA binding

224. A lesion of the left dorsal (parietal) optic radiation produces:

A. Right superior quadrantanopia
B. Right inferior quadrantanopia
C. Right homonymous hemianopia
D. Bitemporal hemianopia
E. Central scotoma

225. Neurofibromatosis primarily affects which cell type?

A. Astrocyte
B. Oligodendrocyte
C. Schwann cell
D. Microglia
E. Ependymal cell

Answer Key & Explanations

Lectures 17–29 (Exam 2 / FA-SA2) · 225 Questions · Single Best Answer

Lecture 17 — Motor System (27 questions)

Q1. About myotatic reflex of the knee, which is correct?

Correct answer: D. Heteronymous = hamstring

Antagonist inhibition during the knee-jerk reflex targets the hamstring (heteronymous inhibition) — the archive answer 'Afferent = Ib fiber' is wrong; the myotatic reflex afferent is Ia (muscle spindle), not Ib (Golgi tendon organ).

✥ *Corrected from the original archive extraction, which had marked A. Afferent = Ib fiber — that was wrong.*

Q2. When dynamic γ-motor neurons activate with α-motor neurons:

Correct answer: E. More Ia impulses than α alone

Alpha-gamma coactivation keeps the spindle taut during contraction, generating MORE Ia impulses than alpha activation alone. The archive answer 'Clonus' is wrong — clonus is an abnormal UMN-lesion finding, unrelated to normal coactivation physiology.

✥ *Corrected from the original archive extraction, which had marked B. Clonus occurs — that was wrong.*

Q3. Which statement about muscle spindles is correct?

Correct answer: B. Bag: static & dynamic; Chain: static

Nuclear bag fibers carry both static and dynamic responses (dynamic bag1 + static bag2); nuclear chain fibers carry mainly static response. The archive answer ('Bag: dynamic; Chain: dynamic') is wrong on both counts.

✥ *Corrected from the original archive extraction, which had marked C. Bag: dynamic; Chain: dynamic — that was wrong.*

Q4. Regarding the knee myotatic reflex, which statement is correct?

Correct answer: D. Heteronymous = hamstring

Same fact as Q1 — heteronymous inhibition targets the hamstring. The archive answer 'Antagonist = quadriceps' is backwards; quadriceps is the agonist/homonymous muscle being tested, not the antagonist.

✥ *Corrected from the original archive extraction, which had marked C. Antagonist = quadriceps — that was wrong.*

Q5. Regarding muscle spindles, which statement is correct?

Correct answer: B. Bag = static & dynamic; Chain = static

Bag fibers carry static AND dynamic responses; chain fibers carry mainly static response.

Q6. When dynamic γ-motor neurons are activated with α-motor neurons:

Correct answer: E. More Ia impulses are generated than with α-activation alone

Same fact as Q2 — coactivation generates MORE Ia impulses than alpha alone, not muscle failure to contract.

✥ *Corrected from the original archive extraction, which had marked C. Muscle won't contract — that was wrong.*

Q7. Which of the following structures is NOT part of the motor system?

Correct answer: D. Cingulate gyrus of cerebrum

The cingulate gyrus is a limbic structure, not a core motor-system component (cerebellum, basal ganglia, extrapyramidal tracts, and the NMJ/muscle all are).

Q8. In the biceps reflex, the antagonist muscle is the:

Correct answer: B. Antagonist = triceps

For the biceps reflex, the antagonist muscle is the triceps.

Q9. Which statement about muscle spindles is correct?

Correct answer: B. Bag: static & dynamic, Chain: static

Same static/dynamic fact as Q3 — chain fibers are mainly STATIC, not dynamic. The archive answer has this backwards.

✥ *Corrected from the original archive extraction, which had marked E. Bag: static & dynamic, Chain: dynamic — that was wrong.*

Q10. Damage to which structure gives lower motor neuron (LMN) signs?

Correct answer: C. Motor nuclei of brainstem

LMN signs arise from damage to the motor neuron cell bodies themselves — brainstem motor nuclei (or spinal anterior horn cells) — not from damage to UMN pathways (crus cerebri, precentral gyrus, internal capsule).

Q11. A patient has left lower facial weakness and the tongue deviates to the left when protruded. The lesion is in the:

Correct answer: B. Right corticobulbar tract

Lower facial weakness (forehead spared) plus tongue deviation to the same side both point to a single UMN (corticobulbar) lesion on the side OPPOSITE the deficits — here, the right corticobulbar tract.

Q12. When α and γ motor neurons co-activate:

Correct answer: E. More Ia impulses are generated than with α-activation alone

Same coactivation fact as Q2/Q6 — more Ia impulses are generated with coactivation than with alpha alone. 'Inhibit Ib afferents' doesn't describe this mechanism at all.

✥ Corrected from the original archive extraction, which had marked A. Inhibit Ib afferents — that was wrong.

Q13. While walking on the beach, a patient suddenly steps on a sharp shell. Which sensory neuron/afferent fiber activates the...

Correct answer: A. A-delta

A-delta fibers carry the fast, sharp pain signal that triggers the flexor withdrawal reflex.

Q14. Given the table comparing static and dynamic responses in muscle spindles, which row is correct?

Correct answer: B. Static: Nuclear chain & bag, Dynamic: Nuclear bag

Static response comes from BOTH chain and (static) bag fibers; dynamic response comes specifically from (dynamic) bag fibers. The archive answer reverses bag and chain's roles entirely.

✥ Corrected from the original archive extraction, which had marked A. Static: Nuclear bag, Dynamic: Nuclear chain — that was wrong.

Q15. Which is NOT a sign of an upper motor neuron (UMN) lesion?

Correct answer: E. Atrophy

Atrophy is NOT a prominent UMN sign — it's the hallmark of LMN lesions (denervation atrophy). Hyperreflexia (the archive's answer) is very much a classic UMN sign, making it a poor choice for 'NOT a sign of UMN lesion.'

✥ Corrected from the original archive extraction, which had marked B. Hyperreflexia — that was wrong.

Q16. Which of the following is NOT a pathological reflex?

Correct answer: C. Cremasteric reflex

The cremasteric reflex is a NORMAL physiological superficial reflex, not pathological. The Babinski sign (the archive's answer) IS classically pathological in adults, making it a poor fit for 'NOT a pathological reflex.'

✥ Corrected from the original archive extraction, which had marked A. Babinski sign — that was wrong.

Q17. A patient has a lesion in the left pyramidal tract for one month. Which finding is NOT expected?

Correct answer: D. Muscular atrophy

At one month post-lesion, the chronic UMN pattern (hypertonia, hyperreflexia, paresis, Babinski) is expected — but muscular atrophy is NOT a prominent UMN finding (that's more a LMN sign).

Q18. Which statement about muscle spindles is correct?

Correct answer: B. Bag = static & dynamic; Chain = static

Bag fibers: static and dynamic. Chain fibers: mainly static.

Q19. Which statement about the motor system is correct?

Correct answer: A. Lateral vestibulospinal tract facilitates extensor muscles.

The lateral vestibulospinal tract facilitates extensor (antigravity) muscle tone — the one true statement among these options.

Q20. Which of the following is not a pathological reflex?

Correct answer: A. Cremasteric

The cremasteric reflex is a normal superficial reflex, not pathological.

Q21. Which of the following is correct regarding the Achilles tendon reflex?

Correct answer: A. Receptor is muscle spindle

Illegible in the original archive. The Achilles reflex receptor is the muscle spindle (like all deep tendon reflexes, despite the name) — not the Golgi tendon organ (C), and the effector is gastrocnemius/calf muscle via the tibial nerve, not hamstring (B); the reflex center is S1-S2, not L1-L2 (D).

✥ Corrected from the original archive extraction, which had marked None. (unmarked/illegible) — that was wrong.

Q22. Which of the following is not related to the motor system?

Correct answer: B. Limbic lobe

The limbic lobe is not a core motor-system structure.

Q23. Which statement about the motor system is correct?

Correct answer: A. Lateral vestibulospinal tract facilitates extensor muscles.

Same options as Q19 — the lateral vestibulospinal tract facilitating extensor muscles is the one true statement. The archive answer here ('mammillary body from dorsal tegmentum of midbrain') is false — the mammillary body is a hypothalamic structure, not a midbrain one.

✥ Corrected from the original archive extraction, which had marked C. Mammillary body comes from the dorsal tegmentum in the Midbrain. — that was wrong.

Q24. According to the Neocortex, what is incorrect?

Correct answer: A. VPL terminates at layer 5.

This is the FALSE statement (matching what's asked). Thalamocortical sensory afferents (like VPL) terminate predominantly in cortical LAYER 4, not layer 5 — layer 5 is where large pyramidal output neurons (e.g., corticospinal tract origin) live.

Q25. Which of the following is correct regarding the Achilles tendon reflex?

Correct answer: C. Afferent and efferent fibers are in the Tibial nerve

Both the afferent (Ia, muscle spindle) and efferent (motor) limbs of the Achilles reflex travel via the tibial nerve.

Q26. The reflexive head-turning toward a sudden sound is mediated predominantly by the:

Correct answer: C. Tectospinal tract

The reflexive head/eye turn toward a sudden sound is mediated by the tectospinal tract (inferior colliculus → tectospinal tract → spinal motor nuclei) — not the vestibulospinal tract, which instead handles postural/antigravity reflexes.

✥ Corrected from the original archive extraction, which had marked A. Vestibulospinal tract — that was wrong.

Q27. A lower motor neuron (LMN) lesion of the facial nerve results in:

Correct answer: C. Entire ipsilateral facial paralysis

An LMN facial nerve lesion (e.g., Bell's palsy) causes COMPLETE ipsilateral facial paralysis (both upper and lower face), unlike a UMN lesion which spares the upper face.

Lecture 18 — Cerebellum (22 questions)

Q28. Which tract affects ataxia in this case?

Archive answer: A. Inferior spinocerebellar tract

This question references a specific case/image not captured in the extracted text — cannot independently verify which tract is implicated without seeing the referenced case.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q29. What is a sign of inferior cerebellar peduncle damage?

Correct answer: A. Ipsilateral ataxia

Inferior cerebellar peduncle damage classically causes ipsilateral ataxia — the direct, expected finding. Cerebellar lesions cause HYPOTONIA, not hypertonia (the archive's answer), making it doubly wrong.

✥ Corrected from the original archive extraction, which had marked E. Hypertonia ipsilateral — that was wrong.

Q30. A 60-year-old woman uses melatonin for insomnia. Which organ excretes melatonin?

Correct answer: E. Pineal gland

The pineal gland produces and secretes melatonin. The archive answer 'Cerebellum' has no role in melatonin synthesis or secretion whatsoever — likely a stray/garbled extraction.

✥ Corrected from the original archive extraction, which had marked C. Cerebellum — that was wrong.

Q31. Which structure does NOT concern the stabilizing (vestibulocerebellar) circuit?

Archive answer: B. Fastigial nucleus

FLAGGED — the fastigial nucleus (the archive's answer) is definitely PART OF the vestibulocerebellar circuit (confirmed by Q43 in this same set), so marking it as excluded is inconsistent. Pontine nuclei and the middle cerebellar peduncle are the classic gateway for the CEREBROcerebellum (corticopontocerebellar pathway) instead, a different functional division — a more likely candidate for 'not part of' the vestibulocerebellar circuit, though I can't rule out lateral dentate nucleus (also cerebrocerebellum-associated) as the intended answer without more source context.

■ *Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.*

Q32. Damage to which structure causes ataxia?

Correct answer: C. Inferior cerebellar peduncle

The inferior cerebellar peduncle carries the bulk of proprioceptive/vestibular afferents crucial for coordinated movement — damage classically causes ataxia.

Q33. The output neuron of the cerebellar cortex is the:

Correct answer: E. Purkinje cell

Purkinje cells are THE sole output neuron of the cerebellar cortex, projecting (via inhibitory GABAergic synapses) to the deep cerebellar nuclei. Basket cells (the archive's answer) are inhibitory INTERNEURONS within the cortex, not its output.

✥ *Corrected from the original archive extraction, which had marked C. Basket cell — that was wrong.*

Q34. Which of the following is not part of the cerebellum?

Archive answer: D. Brainstem

FLAGGED — ambiguous question intent. If asking about direct anatomical connection, the brainstem (the archive's answer) is clearly connected to the cerebellum via the three peduncles, making it a poor 'not part of' answer. If asking about a functionally separate motor-control system, basal ganglia (historically taught as a parallel, non-overlapping circuit from the cerebellum) may be the better answer — I can't be certain which interpretation the question intends.

■ *Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.*

Q35. The inability to perform rapid alternating movements combined with an intention tremor is known as:

Correct answer: A. Dysdiadochokinesia + intention tremor

Dysdiadochokinesia (inability to perform rapid alternating movements) plus intention tremor are both classic cerebellar signs.

Q36. A patient has a resting tremor that improves with movement and a dopamine deficiency. This suggests:

Correct answer: A. Parkinson’s disease

Resting tremor that improves with movement, plus dopamine deficiency, is the classic Parkinson's disease presentation.

Q37. A 72-year-old man has an inability to perform rapid alternating movements and an intention tremor, causing him to fall to the...

Correct answer: A. Dysdiadochokinesia and intention tremor

Same cerebellar sign combination as Q35.

Q38. Which cerebellar deep nucleus is primarily for motor planning?

Correct answer: D. Lateral dentate

The lateral cerebellar hemisphere/dentate nucleus (via the cerebrocerebellum) is associated with motor planning, as opposed to the medial vermis/fastigial nucleus (posture) or intermediate zone (ongoing limb execution).

Q39. Which of the following tracts to the cerebellum does not carry proprioception?

Correct answer: D. Tectocerebellar

Tectocerebellar fibers carry visual/auditory information from the colliculi, not proprioception. The archive answer 'Dorsal spinocerebellar' is wrong — that tract is THE classic proprioceptive pathway to the cerebellum, so it definitely does carry proprioception.

✥ *Corrected from the original archive extraction, which had marked A. Dorsal spinocerebellar 26 — that was wrong.*

Q40. Which of the following tracts can inhibit the nucleus in the cerebellum 4 times per second?

Correct answer: B. Olivocerebellar tract

Climbing fibers (olivocerebellar tract) are well known for their characteristic low firing rate, often cited around 1-10 Hz (commonly quoted as ~4 Hz) — a distinguishing feature versus high-frequency mossy fiber input. The archive answer 'Vestibulocerebellar' doesn't have this specific characteristic firing rate.

✥ *Corrected from the original archive extraction, which had marked D. Vestibulocerebellar tract — that was wrong.*

Q41. A damaged flocculonodular lobe of the cerebellum results in:

Correct answer: C. Gait ataxia

The flocculonodular lobe is the vestibulocerebellum, associated with balance/equilibrium and gait — damage causes gait (truncal) ataxia. Dysdiadochokinesia (the archive's answer) is instead a lateral-hemisphere (cerebrocerebellum) sign related to fine limb coordination.

✦ Corrected from the original archive extraction, which had marked B. Dysdiadochokinesia — that was wrong.

Q42. Fibers from the dentate nucleus of the cerebellum travel to the _____ of the thalamus by passing through the _____.

Correct answer: C. VL / Superior cerebellar peduncle

Dentate nucleus output travels via the superior cerebellar peduncle to the VL nucleus of the thalamus, completing the cerebellar-thalamo-cortical loop.

Q43. The stabilizing circuit (vestibulocerebellum) synapses with which nucleus in the cerebellum?

Correct answer: D. Fastigial nucleus

The fastigial nucleus is the deep cerebellar nucleus associated with the vestibulocerebellum/stabilizing circuit — confirms Q31's archive answer (which excluded fastigial) was inconsistent.

Q44. Which pair represents an excitatory input? 28

Correct answer: E. Mossy fiber - Granular cell

Mossy fibers synapse onto granule cells via an excitatory glutamatergic synapse in the cerebellar glomerulus.

Q45. The AICA (Anterior Inferior Cerebellar Artery) supplies all of the following EXCEPT: 30

Correct answer: D. Crus cerebri

Crus cerebri is a midbrain structure supplied by posterior cerebral artery/basilar perforators, not AICA. The archive answer 'Dorsolateral caudal pons' is wrong — that region IS classic AICA territory (lateral pontine syndrome).

✦ Corrected from the original archive extraction, which had marked A. Dorsolateral caudal pons — that was wrong.

Q46. A patient has symptoms of dysdiadochokinesia and intention tremor. This is indicative of a lesion in the:

Correct answer: A. Cerebellum

Dysdiadochokinesia plus intention tremor localizes to the cerebellum.

Q47. A flocculonodular lobe lesion leads to:

Correct answer: C. Gait ataxia

Same fact as Q41 — flocculonodular lobe lesions cause gait ataxia, not resting tremor (which is a basal ganglia/Parkinson's sign, unrelated to cerebellar pathology).

✦ Corrected from the original archive extraction, which had marked A. Resting tremor 38 — that was wrong.

Q48. Astereognosis (inability to identify an object by touch) is caused by a lesion in the:

Archive answer: B. Primary somatosensory cortex

FLAGGED — astereognosis is more precisely a lesion of the PARIETAL ASSOCIATION cortex (particularly superior parietal lobule) rather than primary somatosensory cortex itself, since primary S1 damage causes basic sensory loss rather than the specific 'can feel but can't identify' agnosia pattern. Given the options provided don't include a specific association-cortex choice, primary somatosensory cortex is the closest available answer, but this is an imprecise fit.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q49. Climbing fibers to the cerebellum originate from the:

Correct answer: B. Inferior olivary nucleus

Climbing fibers originate from the inferior olivary nucleus.

Lecture 19 — Cerebral Cortex (31 questions)

Q50. A 73-year-old woman presents with right arm weakness and left gaze preference after recovery from a cerebrovascular accident....

Correct answer: A. Frontal

Right arm weakness (left corticospinal lesion) plus left gaze preference (left frontal eye field driving unopposed leftward gaze) both localize to the left frontal lobe.

Q51. Which of the following pathways connects Wernicke’s area to Broca’s area?

Correct answer: C. Arcuate fasciculus

The arcuate fasciculus connects Wernicke's and Broca's areas; its damage causes conduction aphasia.

Q52. Which cortical cells provide axons to corpus callosum?

Correct answer: B. External pyramidal (pyramidal cells)

Commissural (corpus callosum) axons classically arise from layer III (external pyramidal) neurons. The archive answer 'Internal pyramidal' (layer V) is instead associated with long PROJECTION fibers (e.g., corticospinal tract to the spinal cord), a different fiber category.

✥ Corrected from the original archive extraction, which had marked D. Internal pyramidal — that was wrong.

Q53. In stroke recovery, a 73-year-old woman has right arm weakness and left gaze preference. Which gyrus is most likely injured?

Correct answer: A. Left posterior middle frontal

Right arm weakness plus left gaze preference both localize to the left posterior middle frontal gyrus (housing the arm motor area and adjacent frontal eye field) — not the cuneus/lingual gyri, which are OCCIPITAL (visual) structures unrelated to this presentation.

✥ Corrected from the original archive extraction, which had marked C. Left cuneus & lingual — that was wrong.

Q54. Which of the following is NOT a feature of Klüver-Bucy syndrome?

Correct answer: A. Lesions of the anterior thalamus

Klüver-Bucy syndrome is classically caused BY bilateral temporal lobe (amygdala) lesions — that's a defining feature, not something to exclude. The archive answer excluded the wrong item; anterior thalamic lesions are NOT a Klüver-Bucy feature, making that the correct 'NOT a feature' answer.

✥ Corrected from the original archive extraction, which had marked E. Bilateral temporal lobe lesion — that was wrong.

Q55. Olfactory information reaches the temporal lobe via:

Correct answer: B. Lateral olfactory stria

Olfactory information reaches primary (piriform) cortex via the lateral olfactory stria — uniquely bypassing the thalamus, unlike every other sensory modality. The archive answer 'Anterior thalamic nucleus' relates to the Papez/limbic circuit, not primary olfactory relay.

✥ Corrected from the original archive extraction, which had marked D. Anterior thalamic nucleus — that was wrong.

Q56. A 38-year-old man has glioblastoma in the green-demarcated brain region. Which deficit occurs?

Archive answer: E. Horner’s syndrome

This question references a specific highlighted brain region in an image not captured in the extracted text — cannot independently verify without seeing which region was demarcated.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q57. A 63-year-old woman has spastic paralysis, right hemianesthesia, and a right facial droop, but no aphasia. The lesion is in the:

Correct answer: E. Posterior limb & genu of left internal capsule

Contralateral weakness + hemianesthesia + facial droop WITHOUT aphasia points to a compact SUBCORTICAL lesion affecting all three modalities together while sparing cortical language areas — the posterior limb and genu of the internal capsule fits perfectly. The archive answer 'Pyramid of medulla (right)' is wrong both anatomically (wouldn't explain the facial or sensory findings) and in laterality (a right pyramid lesion would cause LEFT-sided, not right-sided, deficits).

✥ Corrected from the original archive extraction, which had marked D. Pyramid of medulla (right) — that was wrong.

Q58. Which pathway most likely connects Wernicke’s area to Broca’s area in the left cerebral cortex?

Correct answer: C. Arcuate fasciculus

Same core fact as Q51 — the arcuate fasciculus connects Wernicke's and Broca's areas, not the anterior commissure (which connects olfactory/some temporal structures between hemispheres).

✥ Corrected from the original archive extraction, which had marked A. Anterior commissure — that was wrong.

Q59. A 73-year-old woman presents with right arm weakness and left gaze preference after a cerebrovascular accident. Sensory and...

Correct answer: A. Frontal

Same case as Q50 — right arm weakness plus left gaze preference localizes to the frontal lobe.

Q60. In the same patient from Q3 (right arm weakness, left gaze preference), which gyrus is injured?

Correct answer: A. Left posterior middle frontal

Same case as Q50/Q53 — localizes to the left posterior middle frontal gyrus.

Q61. A 71-year-old man with a left CVA presents with dysprosody, impaired singing, poor judgment, impulsiveness, lethargy, and...

Archive answer: E. Limbic lobe

FLAGGED — dysprosody, impaired singing, poor judgment, impulsiveness, and indifference together most classically describe a FRONTAL LOBE syndrome (personality/executive changes), not limbic lobe specifically, though cingulate/limbic structures do contribute to some apathy-related findings. Moderate confidence the archive's 'Limbic lobe' answer is wrong and should be 'Frontal lobe' instead.

■ *Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.*

Q62. Bilateral destruction of the amygdala causes:

Correct answer: A. Klüver-Bucy syndrome

Bilateral amygdala destruction causes Klüver-Bucy syndrome.

Q63. Which cortical neurons provide axons to the corpus callosum?

Correct answer: B. External pyramidal

Same fact as Q52 — layer III (external pyramidal) neurons provide corpus callosum (commissural) axons.

Q64. Ischemia of the paracentral lobule and cingulate gyrus results from occlusion of the:

Correct answer: B. Callosomarginal artery

The callosomarginal artery (an ACA branch) supplies the paracentral lobule and cingulate gyrus.

Q65. A 71-year-old man presents with contralateral hemineglect, anosognosia, and visuospatial disability, but no aphasia. Which...

Correct answer: B. Parietal

Hemineglect, anosognosia, and visuospatial disability without aphasia is the classic (typically non-dominant/right) parietal lobe syndrome.

Q66. The habenulointerpeduncular tract projects to the:

Correct answer: A. Midbrain

The habenulointerpeduncular tract (fasciculus retroflexus) connects the habenula to the interpeduncular nucleus — a midbrain structure, literally named in the tract itself. The archive answer 'Neurohypophysis' is unrelated; that's reached via different hypothalamic tracts (supraopticohypophyseal, etc.), not this one.

✥ *Corrected from the original archive extraction, which had marked E. Neurohypophysis — that was wrong.*

Q67. A patient has abnormal arm and leg movements but remains fully conscious. This is a:

Correct answer: A. Simple partial seizure

Illegible in the original archive. Abnormal limb movements WITH preserved consciousness is, by definition, a simple partial seizure — complex partial seizures by definition involve impaired consciousness.

✥ *Corrected from the original archive extraction, which had marked None. (unmarked/illegible) — that was wrong.*

Q68. Damage to the paracentral lobule and cingulate gyrus is caused by occlusion of which artery?

Correct answer: A. Callosomarginal artery

Illegible in the original archive. Same fact as Q64 — the callosomarginal artery supplies the paracentral lobule/cingulate gyrus.

✥ *Corrected from the original archive extraction, which had marked None. (unmarked/illegible) — that was wrong.*

Q69. A patient has finger agnosia and dyscalculia. The lesion is in the:

Correct answer: A. Left angular gyrus

Finger agnosia plus dyscalculia (part of the Gerstmann tetrad, with agraphia and left-right disorientation) localizes to the dominant (left) angular gyrus — not the precuneus, a different parietal structure involved in visuospatial/self-processing.

✥ *Corrected from the original archive extraction, which had marked C. Left precuneus — that was wrong.*

Q70. A patient presents with hypersexuality, hyperorality, and placidity. The lesion is in the:

Correct answer: B. Anterior temporal lobe

Hypersexuality, hyperorality, and placidity together describe Klüver-Bucy syndrome, localizing to the (bilateral) anterior temporal lobe.

Q71. An MRI shows a lesion in the right pars opercularis. The patient will likely have what disorder?

Correct answer: E. Dysprosody

A lesion in the RIGHT (non-dominant) pars opercularis — the homolog of Broca's area — classically causes dysprosody/aprosodia (loss of the emotional/melodic tone of speech), mirroring how a LEFT-sided lesion causes true motor (Broca's) aphasia.

Q72. A lesion in the frontal, parietal, and temporal lobes of the non-dominant hemisphere would most likely cause:

Correct answer: D. Dysprosody

Aphasias (Broca's, Wernicke's) are dominant-hemisphere language phenomena. A lesion of the NON-dominant hemisphere would not cause a classic aphasia — it more likely causes dysprosody (the non-dominant-hemisphere analog of a language deficit), consistent with the pattern in Q71.

✥ Corrected from the original archive extraction, which had marked C. Wernicke's Aphasia — that was wrong.

Q73. A patient has right-sided neglect but no aphasia. The lesion is likely in the:

Archive answer: A. Frontal lobe

FLAGGED — hemineglect syndromes are overwhelmingly classically taught as a PARIETAL lobe phenomenon (matching Q65's correctly-identified pattern), not frontal, though frontal attentional networks do contribute to neglect in more advanced literature. Moderate-high confidence the archive's 'Frontal' answer should be 'Parietal' for basic-course-level expectations.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q74. A patient has abnormal movement in their right arm and does not lose consciousness. What is the best description of the...

Correct answer: A. Simple partial seizure

By definition, if consciousness is NOT impaired, this is a simple partial seizure, not complex partial (which requires impaired consciousness by definition). The archive answer has this backwards.

✥ Corrected from the original archive extraction, which had marked B. Complex partial seizure — that was wrong.

Q75. A patient has abnormal movement in their right arm but loses consciousness. What is the best description of the patient's...

Correct answer: B. Complex partial seizure

The mirror-image error from Q74: if consciousness IS lost, this is a complex partial seizure, not simple partial (which by definition preserves consciousness).

✥ Corrected from the original archive extraction, which had marked A. Simple partial seizure — that was wrong.

Q76. A lesion in the left cerebral cortex affects the corticobulbar fibers. What is true according to the lesion?

Correct answer: C. Right weakness in closing mouth tightly

A UMN (corticobulbar) facial lesion causes CONTRALATERAL LOWER face weakness only — eyelid closure (upper face) is spared due to bilateral cortical innervation. The archive answer ('inability to close eyelids') describes exactly the finding that should be SPARED, not present, in a UMN lesion — the correct finding is lower-face weakness (e.g., can't close the mouth tightly).

✥ Corrected from the original archive extraction, which had marked B. Right inability to close eyelids — that was wrong.

Q77. A patient exhibits hyperorality and hypersexuality. This is characteristic of:

Correct answer: A. Klüver-Bucy syndrome

Hyperorality plus hypersexuality is classic Klüver-Bucy syndrome.

Q78. The fiber tract connecting Broca's and Wernicke's areas is the:

Correct answer: C. Arcuate fasciculus

The arcuate fasciculus connects Broca's and Wernicke's areas.

Q79. A patient understands language but cannot speak fluently. The lesion is at:

Correct answer: C. Broca's area

Preserved comprehension with non-fluent speech is classic Broca's (motor) aphasia.

Q80. Alexia without agraphia (pure word blindness) is caused by a lesion in the:

Correct answer: C. Splenium of the corpus callosum

Alexia without agraphia results from a splenium of the corpus callosum lesion (disconnecting the intact right visual cortex from the left language areas).

Lecture 20 — Basal Ganglia (11 questions)

Q81. The globus pallidus projects to the thalamus via:

Correct answer: A. Ansa lenticularis

The globus pallidus projects to the thalamus via the ansa lenticularis (and lenticular fasciculus).

Q82. Which fibers are excitatory in the basal ganglia pathway?

Correct answer: C. STN → GPi

The subthalamic nucleus (STN) is the one excitatory (glutamatergic) node in the basal ganglia circuit — STN→GPi is excitatory, while all the other listed connections are inhibitory GABAergic.

Q83. A 25-year-old woman collapses with tonic stiffening for 30 seconds, followed by jerking for 2 minutes. What is the diagnosis?

Correct answer: C. Tonic-clonic seizure

Tonic stiffening followed by clonic jerking is the classic generalized tonic-clonic seizure pattern.

Q84. A woman suddenly collapses with tonic stiffening followed by clonic jerks. The diagnosis is:

Correct answer: C. Tonic-clonic seizure

Identical clinical picture to Q83 (tonic stiffening + clonic jerks) — should also be tonic-clonic seizure, not absence seizure (which shows brief staring with no tonic/clonic movement at all, a completely different presentation).

✦ Corrected from the original archive extraction, which had marked D. Absence seizure — that was wrong.

Q85. Which basal ganglia structure is in the midbrain?

Correct answer: E. Substantia nigra (pars reticulata)

Substantia nigra (both pars compacta and pars reticulata) is unambiguously a midbrain structure.

Q86. A 65-year-old man has a resting tremor, bradykinesia, shuffling gait, and decreased dopamine on a PET scan. The diagnosis is:

Correct answer: D. Parkinson’s disease

Resting tremor, bradykinesia, shuffling gait, and decreased dopamine on PET is classic Parkinson's disease.

Q87. What is the most common cause of hemiballismus–hemichorea?

Correct answer: A. Stroke

Hemiballismus-hemichorea is most commonly caused by a lacunar stroke of the contralateral subthalamic nucleus.

Q88. A patient with uncontrolled movements has atrophy of the basal ganglia. Her mother had similar symptoms. What is her condition?

Correct answer: D. Huntington's Disease

Chorea, basal ganglia (caudate) atrophy, and an autosomal-dominant family history together describe Huntington's disease.

Q89. A patient has Parkinson's disease. What is the effect on the basal ganglia pathways?

Correct answer: B. Underactivity of the direct pathway, overactivity of the indirect pathway

In Parkinson's (dopamine deficiency), loss of D1-mediated facilitation makes the direct pathway underactive, while loss of D2-mediated inhibition makes the indirect pathway overactive (disinhibited).

Q90. Which of the following indicates Parkinson's disease more specifically than other movement disorders?

Correct answer: A. Unilateral resting tremor

Parkinson's classically starts unilaterally with a resting tremor — chorea points to Huntington's, intention tremor to cerebellar disease, and MRI is typically nonspecific/normal in Parkinson's (a clinical diagnosis).

Q91. Which supraspinal site is the key initiator of descending analgesia recruited by opioids?

Correct answer: C. Periaqueductal gray (PAG)

The periaqueductal gray (PAG) is the classic supraspinal site initiating opioid-mediated descending analgesia.

Lecture 21 — Hypothalamus & Epithalamus (10 questions)

Q92. A 30-year-old man had mammillary body damage. Which symptom is found?

Correct answer: D. Loss of recent memory

Mammillary body damage (part of the Papez circuit, classically injured in Wernicke-Korsakoff syndrome) causes loss of recent memory (anterograde amnesia) — not hyperorality, which is a Klüver-Bucy/temporal lobe sign, unrelated to the mammillary body.

✦ Corrected from the original archive extraction, which had marked A. Hyperorality — that was wrong.

Q93. A 30-year-old man with mammillary body damage would present with:

Correct answer: D. Loss of recent memory

Same fact as Q92 — mammillary body damage causes loss of recent memory.

Q94. Which hypothalamic nucleus lies in the tuberal region?

Correct answer: A. Arcuate

The arcuate nucleus lies in the tuberal (middle) region of the hypothalamus, along with the ventromedial and dorsomedial nuclei.

Q95. A 35-year-old man presents with constant hunger, weight loss, and poor concentration. Which hypothalamic nucleus is damaged?

Archive answer: D. Paraventricular

FLAGGED — constant hunger, weight loss, and poor concentration suggest a hunger/appetite-regulation problem. The arcuate nucleus is the modern primary integration center for hunger signals (NPY/AgRP and POMC/CART neurons), making it a strong candidate over paraventricular nucleus (the archive's answer, more classically tied to releasing-hormone and oxytocin/vasopressin synthesis, not primary appetite regulation) — though the classic 'ventromedial=satiety/lateral=hunger' centers aren't offered as options here, adding some uncertainty.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q96. What increases melatonin synthesis?

Archive answer: B. Inhibited paraventricular nucleus

FLAGGED — the archive answer 'Inhibited paraventricular nucleus' appears backwards: PVN sympathetic output needs to be enhanced (disinhibited), not inhibited, to drive melatonin synthesis. The pathway is: darkness → SCN activity falls → less SCN inhibition of PVN → PVN sympathetic output rises → superior cervical ganglion stimulated → melatonin increases. 'Inhibited SCN' is the more classic textbook phrasing for the initiating event, though 'stimulated superior cervical ganglion' is also defensible as the more proximate cause.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q97. Which nucleus lies in the supraoptic region of the hypothalamus?

Archive answer: E. Dorsolateral

FLAGGED — 'Dorsolateral' (the archive's answer) is not a standard named hypothalamic nucleus in typical anterior/supraoptic classification, and 'Centromedian' is actually a THALAMIC nucleus (a likely category error in the options). Paraventricular nucleus is a well-established supraoptic-region nucleus and the more defensible answer among these choices.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q98. Jet lag is due to dysregulation of the:

Correct answer: B. Suprachiasmatic nucleus

Jet lag results from the suprachiasmatic nucleus (master circadian clock) being out of sync with the new external light-dark cycle.

Q99. Which statement about the functions of hypothalamic nuclei and their tracts is correct?

Correct answer: C. The posterior hypothalamic nucleus sends sympathetic signals to the spinal cord (T1-L2) via the hypothalamospinal tract.

The posterior hypothalamic nucleus does send sympathetic signals to the spinal cord (T1-L2) via the hypothalamospinal tract — a true statement. The archive answer is false as worded: the mammillothalamic tract connects the mammillary body to the ANTERIOR thalamic nucleus, not the dorsomedial nucleus (confirmed by Q114's correctly-identified Papez circuit fact).

✦ Corrected from the original archive extraction, which had marked B. The Mammillothalamic tract connects the dorsomedial nuclei and the mammillary body. — that was wrong.

Q100. What increases the synthesis of melatonin?

Correct answer: A. Inhibited suprachiasmatic nucleus

Same melatonin pathway as Q96 — inhibited SCN (representing darkness/reduced light input) is the initiating event allowing melatonin to rise. 'Stimulation of the optic tract' (the archive's answer) implies light exposure, which should SUPPRESS, not increase, melatonin — backwards.

✥ Corrected from the original archive extraction, which had marked C. Stimulation of optic tract — that was wrong.

Q101. The heat gain center in the hypothalamus is located in the:

Correct answer: B. Posterior nucleus

The posterior hypothalamic nucleus is the heat-gain/conservation center (shivering, vasoconstriction) — the anterior nucleus is the heat-loss center instead.

Lecture 22 — Thalamus (14 questions)

Q102. Which thalamic nucleus receives input from contralateral lateral spinothalamic tract?

Correct answer: E. Ventral posterolateral nucleus

VPL (ventral posterolateral) is the classic thalamic relay for body pain/temperature/touch via the spinothalamic and DCML pathways.

Q103. Which sensation bypasses the thalamus before reaching the cortex?

Correct answer: E. Olfaction

Olfaction is the ONE sensory modality that bypasses the thalamus entirely en route to primary cortex. The archive answer 'Vision' is wrong — vision definitely relays through the thalamus (lateral geniculate nucleus).

✥ Corrected from the original archive extraction, which had marked A. Vision — that was wrong.

Q104. Which sensation does not synapse in the thalamus before reaching the cortex?

Correct answer: E. Olfaction

Same fact as Q103 — olfaction bypasses the thalamus; confirms Q103's 'Vision' answer was wrong.

Q105. Which thalamic nucleus connects to the solitariothalamic tract? 6

Correct answer: E. Ventral posteromedial nucleus

VPM is the thalamic relay for the solitariothalamic (taste) tract.

Q106. Which thalamic nuclei influence activity of the motor cortex?

Correct answer: B. Ventral anterior & ventral lateral nuclei

VA (ventral anterior) and VL (ventral lateral) are the classic thalamic relay nuclei for motor function — VA/VL receive basal ganglia/cerebellar input respectively and project to motor/premotor cortex. The archive answer 'Lateral geniculate & reticular' is wrong — lateral geniculate is purely visual, unrelated to motor cortex.

✥ Corrected from the original archive extraction, which had marked E. Lateral geniculate & reticular nuclei — that was wrong.

Q107. Which thalamic nucleus receives input from the contralateral lateral spinothalamic tract?

Correct answer: E. Ventral posterolateral nucleus

Same fact as Q102 — VPL receives the contralateral spinothalamic tract, not mediodorsal (which relates to limbic/prefrontal connections instead).

✥ Corrected from the original archive extraction, which had marked B. Mediodorsal nucleus 7 — that was wrong.

Q108. Which sensation bypasses the thalamus before reaching the cortex?

Correct answer: E. Smell

Smell (olfaction) bypasses the thalamus; taste (the archive's answer) actually DOES relay through the thalamus, specifically VPM.

✥ Corrected from the original archive extraction, which had marked C. Taste — that was wrong.

Q109. Which thalamic nuclei influence the motor cortex?

Correct answer: B. Ventral anterior & ventral lateral

Same fact as Q106 — VA & VL influence the motor cortex, not dorsomedial & lateral dorsal nuclei (more associative/limbic in function).

✥ Corrected from the original archive extraction, which had marked A. Dorsomedial & lateral dorsal — that was wrong.

Q110. The thalamic nucleus receiving input from the lateral spinothalamic tract is the:

Correct answer: E. Ventral posterolateral

Same fact as Q102/Q107 — VPL receives the lateral spinothalamic tract, not mediodorsal.

✥ Corrected from the original archive extraction, which had marked B. Mediodorsal — that was wrong.

Q111. The thalamic nucleus connected to the solitariothalamic tract is the:

Correct answer: E. Ventral posteromedial

Same fact as Q105 — VPM connects to the solitariothalamic (taste) tract, not centromedian (an intralaminar nucleus tied to arousal/pain modulation more broadly).

✥ Corrected from the original archive extraction, which had marked A. Centromedian — that was wrong.

Q112. Which of the following is correct regarding the touch sensation pathway?

Archive answer: C. The final neuron projects to the primary somatosensory area.

FLAGGED — both A and C are defensible true statements about the touch pathway depending on exact neuron-counting convention. Using the standard 4-neuron-order framework (1st=DRG, 2nd=cuneate/gracile nucleus, 3rd=VPL thalamus, 4th=cortex), option A ('third-order neuron synapses in the thalamus') is the more precisely worded true statement, though C ('final neuron projects to primary somatosensory area') isn't clearly false either.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q113. The gustatory pathway includes all of the following EXCEPT:

Correct answer: E. Trapezoid body

The trapezoid body is part of the AUDITORY pathway, not the gustatory pathway — correctly excluded.

Q114. Within the Papez circuit, damage to which thalamic nucleus impairs recent memory?

Correct answer: C. Anterior nucleus

The anterior thalamic nucleus is the Papez-circuit relay (mammillothalamic tract input, cingulate gyrus output) — damage impairs memory consolidation.

Q115. A lesion of the anterior thalamic nucleus causes:

Correct answer: B. Anterograde amnesia

Same Papez-circuit fact as Q114 — anterior thalamic nucleus damage causes anterograde amnesia, not dysgeusia (which would relate to VPM/gustatory pathway damage, an unrelated nucleus).

✥ Corrected from the original archive extraction, which had marked C. Dysgeusia — that was wrong.

Lecture 23 — Limbic System & Olfaction (12 questions)

Q116. What is the relationship of the hippocampus to the inferior horn of the lateral ventricle?

Correct answer: A. In floor

The hippocampus forms a bulge in the floor of the inferior (temporal) horn of the lateral ventricle.

Q117. A 39-year-old woman has an eating disorder, thirst, and fluctuating temperature. Which structure is dysfunctional?

Correct answer: C. Hypothalamus

Eating disorder, thirst, and temperature dysregulation together are classic HYPOTHALAMUS dysfunction (hypothalamus regulates all three homeostatic functions) — the hippocampus (the archive's answer) has no role in any of them; it's a memory-formation structure.

✥ Corrected from the original archive extraction, which had marked B. Hippocampus — that was wrong.

Q118. Which is NOT a major bidirectional hypothalamic pathway?

Archive answer: C. Dorsal longitudinal fasciculus → PAG

FLAGGED — the retinohypothalamic tract is classically a UNIDIRECTIONAL pathway (retina → SCN only), making it the most likely 'NOT bidirectional' answer. The archive answer (dorsal longitudinal fasciculus → PAG) is generally considered a genuinely bidirectional hypothalamic-brainstem connection in most texts.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q119. A 39-year-old woman has an eating disorder, thirst, and temperature dysregulation. This indicates dysfunction of the:

Correct answer: C. Hypothalamus

Same triad as Q117 — eating/thirst/temperature dysregulation indicates hypothalamic dysfunction; confirms Q117's 'Hippocampus' answer was wrong.

Q120. The conversion of short-term to long-term memory occurs in the:

Correct answer: E. Hippocampus

Short-term to long-term memory conversion (consolidation) occurs in the hippocampus.

Q121. Sara feels fear and has an increased heart rate after hearing a sound at night. Which brain part is most active?

Correct answer: D. Amygdala

Fear and the associated autonomic response (increased heart rate) is classic amygdala activation.

Q122. A 65-year-old alcoholic presents with severe memory impairment. Atrophy is expected in which structure?

Correct answer: E. Mammillary body

Chronic alcoholism with severe memory impairment classically shows mammillary body atrophy (Wernicke-Korsakoff syndrome).

Q123. A patient has an eating disorder, severe thirst, and a wildly varying body temperature. Which structure is most likely...

Correct answer: E. Hypothalamus

Same eating/thirst/temperature triad as Q117/Q119 — this indicates hypothalamic dysfunction, not hippocampus (the archive's answer, repeating the same error pattern for the third time in this question set).

✥ Corrected from the original archive extraction, which had marked D. Hippocampus — that was wrong.

Q124. An alcoholic patient has a memory deficit. Which structure is most likely damaged?

Correct answer: B. Mammillary Body

Same Wernicke-Korsakoff fact as Q122 — mammillary body damage from chronic alcoholism.

Q125. Which of the following is correct about the fornix?

Correct answer: B. Postcommissural fornix connects to the mammillary body.

Illegible in the original archive. The postcommissural fornix carries hippocampal output to the mammillary body (part of the Papez circuit) — a true statement. The precommissural fornix instead projects mainly to the septal area (not directly to the anterior thalamic nucleus, ruling out option A); the fornix originates from the hippocampus, not the amygdala (ruling out C).

✥ Corrected from the original archive extraction, which had marked None. (unmarked/illegible) — that was wrong.

Q126. An alcoholic with a memory deficit most likely has damage to the:

Correct answer: A. Mammillary Body

Alcoholic memory deficit classically implicates the mammillary body (Wernicke-Korsakoff).

Q127. Which stem cell type regenerates sensory receptor cells in the tongue and olfactory epithelium?

Correct answer: C. Basal cells

Basal cells are the shared regenerative stem cell population for both taste buds and olfactory receptor neurons. Satellite cells (the archive's answer) instead surround neuron cell bodies within PNS ganglia — an unrelated function.

✥ Corrected from the original archive extraction, which had marked A. Satellite cells — that was wrong.

Lecture 24 — Autonomic Nervous System I (Structures) (10 questions)

Q128. Throughout the autonomic nervous system, preganglionic fibers release neurotransmitter that binds to which receptor? 3

Correct answer: A. N 2

All preganglionic autonomic fibers (sympathetic and parasympathetic alike) release ACh acting on nicotinic (N2) receptors at the ganglionic synapse.

Q129. Which is NOT a feature of Horner's syndrome?

Archive answer: D. Enophthalmos

FLAGGED — 'apparent' enophthalmos (from ptosis narrowing the palpebral fissure) is commonly included as an associated Horner's finding in many texts, making 'Enophthalmos' (the archive's answer) a debatable choice for 'NOT a feature.' Salivation is primarily under PARASYMPATHETIC control, so 'no saliva secretion' is not something sympathetic (Horner's) disruption would classically cause — a stronger candidate for the true exception.

■ *Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.*

Q130. A lesion in which location can present with Horner's syndrome?

Correct answer: E. Sympathetic chain T1–T4 ■ MED34

A lesion anywhere along the sympathetic pathway to the eye — including the preganglionic sympathetic chain at T1-T4 — can produce Horner's syndrome.

Q131. Which of the following nerves carries parasympathetic fibers?

Correct answer: D. Pelvic splanchnic

All the named splanchnic nerves (greater, lesser, least, lumbar) are SYMPATHETIC except the pelvic splanchnic nerve, which is uniquely parasympathetic (S2-S4 outflow). The archive answer 'Lesser splanchnic' is definitely sympathetic.

✥ *Corrected from the original archive extraction, which had marked C. Lesser splanchnic — that was wrong.*

Q132. In the autonomic nervous system, preganglionic fibers release a neurotransmitter that binds to:

Correct answer: A. N2 receptors

Same fact as Q128 — preganglionic fibers act on nicotinic (N2) receptors.

Q133. Which of the following does NOT form the cardiac plexus?

Correct answer: E. Greater splanchnic nerve

The greater splanchnic nerve supplies ABDOMINAL viscera (via the celiac ganglion) — it has no role in the cardiac plexus. The archive answer 'Superior cardiac branch' is wrong; that branch genuinely IS part of the cardiac plexus.

✥ *Corrected from the original archive extraction, which had marked A. Superior cardiac branch — that was wrong.*

Q134. Which sympathetic/parasympathetic action is correct?

Correct answer: D. Bladder: Symp relaxes, Para contracts

Sympathetic relaxes the detrusor and contracts the internal sphincter (storage); parasympathetic contracts the detrusor and relaxes the sphincter (voiding) — the one correctly-paired statement among these options.

Q135. Which pairing of preganglionic neurons and their fibers is incorrect?

Correct answer: E. T1–T4 → Superior cardiac branch

This pairing, taken as testing whether preganglionic fibers directly form the cardiac branch (they don't — they first synapse in the cervical sympathetic ganglion, and postganglionic fibers form the actual branch), is the intended 'incorrect' pairing among an otherwise accurate list.

Q136. The primary source of circulating epinephrine is:

Correct answer: C. Adrenal medullary chromaffin cells

Adrenal medullary chromaffin cells (a modified sympathetic ganglion) are the primary source of circulating epinephrine.

Q137. Which of the following is a parasympathetic fiber?

Correct answer: D. Pelvic splanchnic nerve

Same fact as Q131 — pelvic splanchnic is the one parasympathetic splanchnic nerve. The archive answer 'Lumbar splanchnic' is sympathetic.

✥ *Corrected from the original archive extraction, which had marked B. Lumbar splanchnic nerve — that was wrong.*

Lecture 25 — Autonomic Nervous System II (Functions) (4 questions)

Q138. A patient has wild mushroom poisoning (muscarinic). Expected findings are:

Archive answer: C. Increased tears, bradycardia, relaxed urinary sphincter 20

FLAGGED — muscarinic (cholinergic) toxicity causes increased secretions (both salivation AND tears via 'SLUDGE'), bradycardia, and a relaxed urinary sphincter. Options A and C both correctly pair bradycardia with a relaxed sphincter but differ only in which secretion (salivation vs. tears) is named — both increase in reality, so either could be defensible depending on which single symptom the question intends to highlight.

■ *Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.*

Q139. Which enzyme converts norepinephrine to epinephrine in the adrenal medulla?

Correct answer: C. Phenylethanolamine N-methyltransferase (PNMT)

PNMT (phenylethanolamine N-methyltransferase) converts norepinephrine to epinephrine in the adrenal medulla. 'Epinephrine synthase' (the archive's answer) is not a real, standard enzyme name.

✦ *Corrected from the original archive extraction, which had marked D. Epinephrine synthase — that was wrong.*

Q140. Norepinephrine for arousal and alertness is synthesized mainly in the:

Correct answer: C. Locus coeruleus

The locus coeruleus is the primary noradrenergic nucleus driving CNS arousal/alertness.

Q141. The enzyme that converts norepinephrine (NE) to epinephrine (Epi) is:

Correct answer: C. PNMT

Same fact as Q139 — PNMT converts NE to epinephrine. COMT (the archive's answer) is instead involved in catecholamine degradation/inactivation, not synthesis.

✦ *Corrected from the original archive extraction, which had marked A. COMT — that was wrong.*

Lecture 26 — EEG & Sleep Physiology (11 questions)

Q142. What are the characteristics of an EEG α-rhythm?

Correct answer: C. Disappears with eye opening

Alpha rhythm (8-13 Hz, not 20-30 Hz which is beta range) is associated with relaxed wakefulness and is classically defined by its disappearance ('alpha blocking') when the eyes open — not deep sleep, which shows slow delta waves instead.

✦ *Corrected from the original archive extraction, which had marked A. Deep sleep — that was wrong.*

Q143. A 10-year-old boy has a generalized seizure with momentary loss of consciousness. His EEG shows 2.5–3 Hz generalized...

Correct answer: B. Absence

Brief loss of consciousness with 2.5-3 Hz generalized spike-wave discharges in a child is the classic absence seizure pattern.

Q144. A 35-year-old man has obstructive sleep apnea. Which EEG pattern occurs during REM sleep?

Correct answer: D. Low amplitude, high frequency

REM sleep EEG is classically low amplitude, high frequency (desynchronized, resembling wakefulness — 'paradoxical sleep'). K complexes and spindles (the archive's answer) are Stage 2 NREM features, not REM.

✦ *Corrected from the original archive extraction, which had marked E. K complexes & spindles — that was wrong.*

Q145. A 35-year-old man transitions from non-REM sleep to wakefulness. Which neurotransmitter change occurs?

Correct answer: D. ↑NE, ↑5-HT, ↑ACh, ↑Histamine, ↓GABA

Transitioning to wakefulness should show INCREASED NE, serotonin, ACh, and histamine, with DECREASED GABA. The archive answer has several of these directions reversed (decreased NE and serotonin, increased GABA) — essentially backwards from the true arousal pattern.

✦ *Corrected from the original archive extraction, which had marked C. ↓NE, ↓5-HT, ↑ACh, ↑Histamine, ↑GABA — that was wrong.*

Q146. The EEG α-rhythm is characterized by:

Correct answer: C. Disappearance when eyes open

Alpha rhythm disappears with eye opening — confirms Q142's archive answer was wrong.

Q147. A 10-year-old boy has brief losses of consciousness. His EEG shows 2.5–3 Hz spike-wave discharges. What is the diagnosis? 16

Correct answer: B. Absence seizures

Same absence seizure pattern as Q143.

Q148. The EEG in REM sleep shows:

Correct answer: D. Low amplitude, high frequency

Same fact as Q144 — REM sleep EEG is low amplitude, high frequency, not high amplitude/high frequency (a combination that doesn't match any standard sleep stage well).

✦ Corrected from the original archive extraction, which had marked A. High amplitude, high frequency — that was wrong.

Q149. What are the changes in neurotransmitters when transitioning from NREM sleep to waking?

Correct answer: B. Locus Coeruleus ↑NE & SE, Raphe nuclei ↓ACh

Transitioning toward wakefulness involves increased noradrenergic (locus coeruleus) and serotonergic (raphe) tone — the archive answer has the correct increasing direction, even if the exact neurotransmitter-to-nucleus phrasing is imprecise.

Q150. Which brainwave pattern occurs in deep sleep (Stage N3)?

Correct answer: E. Delta

Deep sleep (Stage N3, slow-wave sleep) is characterized by delta waves — the slowest, highest-amplitude band. Gamma (the archive's answer) is the FASTEST frequency band, associated with high alertness — the opposite of deep sleep.

✦ Corrected from the original archive extraction, which had marked A. Gamma — that was wrong.

Q151. The EEG waveform in Stage 4 (N3) sleep is characterized by:

Correct answer: B. High amplitude, low frequency

Stage 4 (N3) shows high-amplitude, low-frequency delta activity — confirms Q150's 'Gamma' answer was wrong.

Q152. Which statement about sleep stages is incorrect?

Correct answer: E. REM sleep: night terror ■ MED32

This is the FALSE statement (matching what's asked). Night terrors are a classic NREM (Stage 3/4) parasomnia, not a REM phenomenon — nightmares, by contrast, occur during REM.

Lecture 27 — Reflex Mechanism (6 questions)

Q153. Which receptor mediates corneal reflex sensation?

Correct answer: C. Free nerve endings

The cornea is densely innervated by free nerve endings (CN V1) — it lacks specialized encapsulated receptors like Pacinian corpuscles (the archive's answer).

✦ Corrected from the original archive extraction, which had marked D. Pacinian corpuscles — that was wrong.

Q154. The sensory receptor for the corneal reflex is the:

Correct answer: C. Free nerve endings

Same fact as Q153 — free nerve endings mediate corneal sensation, not Ruffini endings (an encapsulated receptor type not found in the cornea).

✦ Corrected from the original archive extraction, which had marked B. Ruffini endings — that was wrong.

Q155. The cause of the rebound phenomenon (a cerebellar sign) is:

Correct answer: A. No stretch reflex 14

The rebound phenomenon (Holmes sign) reflects impaired rapid antagonist-muscle checking via the stretch reflex, normally coordinated by the cerebellum.

Q156. The Glasgow Coma Scale (GCS) consists of assessing:

Correct answer: A. Motor, verbal responses, eye opening

The GCS assesses eye opening, verbal response, and motor response.

Q157. In the pupillary light reflex, sensory input is via CN ___ and motor output is via CN ___.

Correct answer: A. II, III

The pupillary light reflex uses CN II (afferent, optic nerve) and CN III (efferent, oculomotor/Edinger-Westphal). The archive answer 'V, VII' is the CORNEAL reflex's afferent/efferent pairing instead — a classic mix-up between these two different reflexes.

✦ Corrected from the original archive extraction, which had marked D. V, VII — that was wrong.

Q158. The gag reflex is mediated by which brainstem level?

Correct answer: C. Medulla oblongata

The gag reflex (CN IX afferent, CN X efferent) is mediated at the medulla oblongata level.

Lecture 28 — Blood Supply of Brain & Meninges (31 questions)

Q159. You are caring for a patient who has suffered a stroke and has issues comprehending spoken language (Aphasia). Branches of...

Correct answer: B. MCA

MCA supplies the classic dominant-hemisphere language areas (Broca's, Wernicke's) — occlusion is the classic cause of aphasia.

Q160. Cerebrospinal fluid communicates with the subarachnoid space via which structure?

Archive answer: C. Subarachnoid granulations

FLAGGED — arachnoid granulations are where CSF is REABSORBED from the subarachnoid space into venous blood (an outflow/reabsorption point), not where CSF first communicates INTO the subarachnoid space from the ventricles. The foramina of Luschka and Magendie (located at the 4th ventricle) are the actual connection point, making '4th ventricle' (the more likely intended answer) a better fit for 'communicates with the subarachnoid space.'

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q161. A 55-year-old male, post-stroke, has a shaded lesion in the lower pons (autopsy diagram). The embolus is in which artery?

Correct answer: E. Basilar artery

A lower pons lesion is consistent with basilar artery territory (the parent vessel supplying the whole pons via its branches).

Q162. Which artery is most likely involved in this lesion?

Correct answer: A. Basilar artery

Same image-referenced pontine lesion as Q161 — basilar artery.

Q163. Regarding cerebral cortex blood supply, which is true?

Correct answer: B. ACA: contralateral leg and micturition

ACA supplies the contralateral leg and the micturition control area (paracentral lobule) — the one clearly true statement; the other options either reverse laterality or misattribute territory (e.g., MCA does not supply the leg).

Q164. Superior sagittal sinus is located between:

Correct answer: B. Endosteal dura & meningeal dura

The superior sagittal sinus runs within a split of the dura mater itself, between the outer (endosteal) and inner (meningeal) dural layers.

Q165. Which statement about strokes is false?

Archive answer: A. Stroke = occlusion/rupture

FLAGGED — the archive's marked answer ('Stroke = occlusion/rupture') is actually a TRUE statement (the correct standard definition, confirmed by Q167), so it cannot be the false one being asked for. Multiple other options are also false as written (B: TIA does NOT last 2-3 weeks — it resolves within 24 hours; C and D also misattribute ischemic vs. hemorrhagic mechanisms), making it hard to identify a single intended 'the' false statement without more context — TIA duration (B) is the most likely single intended answer given it's the most specific, commonly-tested individual fact.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q166. A 65-year-old male was admitted after a stroke and later died. The autopsy cross-section of his pons shows a dark lesion. The...

Correct answer: E. Basilar artery

A pontine lesion should implicate the basilar artery, which supplies the pons — not PICA (the archive's answer), which supplies the medulla and inferior cerebellum, a different territory.

✥ Corrected from the original archive extraction, which had marked C. PICA — that was wrong.

Q167. Which of the following statements about stroke is true?

Correct answer: E. Stroke = neurological injury from occlusion or rupture of cerebral vessels

'Stroke = neurological injury from occlusion or rupture of cerebral vessels' is the one TRUE statement (the standard definition). The archive answer ('Ischemic stroke = rupture of vessels') is false — ischemic stroke is caused by OCCLUSION, not rupture (rupture describes hemorrhagic stroke).

✦ Corrected from the original archive extraction, which had marked B. Ischemic stroke = rupture of brain blood vessels — that was wrong.

Q168. An occlusion of the right ACA (anterior cerebral artery) will cause deficits in:

Correct answer: B. Motor & sensory in left lower limbs

Right ACA occlusion causes contralateral (left) leg motor/sensory deficits.

Q169. A 46-year-old man presents with apathy, severe right leg weakness greater than arm/face weakness, and urinary incontinence....

Correct answer: E. Left ACA

Right-sided leg-predominant weakness plus urinary incontinence and apathy (paracentral lobule/medial frontal territory) is contralateral to the lesion — meaning the LEFT ACA is occluded, not the right (the archive's answer reverses the laterality).

✦ Corrected from the original archive extraction, which had marked D. Right ACA — that was wrong.

Q170. Regarding cortical blood supply, which statement is true?

Correct answer: C. ACA → contralateral leg & micturition

ACA supplies the contralateral leg and micturition center — the correct pairing. The archive answer incorrectly includes 'leg' in MCA's territory; MCA does not significantly supply the leg (that's ACA/medial cortex territory).

✦ Corrected from the original archive extraction, which had marked E. MCA → contralateral face, arm, leg, speech areas — that was wrong.

Q171. Which artery is closely related to the primary visual area?

Correct answer: A. Calcarine artery

The calcarine artery (a PCA branch) supplies the primary visual cortex.

Q172. An MRI shows a homogeneous pontine cistern mass. Which vessel is most affected?

Archive answer: E. Posterior inferior cerebellar artery

FLAGGED — a pontine cistern mass most directly implicates the basilar artery, which runs through this exact CSF space along the ventral pons. PICA (the archive's answer) is more associated with the medulla/lateral cerebellum, a somewhat different territory, though it does pass nearby.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q173. Aphasia due to artery occlusion is most often from the:

Correct answer: B. MCA

Aphasia (language area damage) is classically from MCA occlusion, confirmed by Q159. The archive answer 'Anterior choroidal artery' supplies deep structures (internal capsule, optic tract), not primarily cortical language areas.

✦ Corrected from the original archive extraction, which had marked E. Anterior choroidal artery — that was wrong.

Q174. Right ACA occlusion causes a deficit in:

Correct answer: B. Left leg motor/sensory

Same fact as Q168 — right ACA occlusion causes left leg deficits.

Q175. A 58-year-old male presents with right leg weakness greater than arm/face weakness, plus apathy. Which artery is occluded?

Correct answer: E. Left ACA

Same clinical picture as Q169 — right-sided leg-predominant weakness localizes to the LEFT ACA; confirms Q169's 'Right ACA' answer was wrong.

Q176. A 55-year-old man has a lesion in the pons found on autopsy. The embolus was likely in the:

Correct answer: E. Basilar artery

Same pontine lesion fact as Q161 — basilar artery.

Q177. The artery most related to the primary visual area is the:

Correct answer: A. Calcarine artery

Same fact as Q171 — the calcarine artery supplies primary visual cortex, not the callosomarginal artery (the archive's answer, which supplies the paracentral lobule/cingulate area instead).

✦ Corrected from the original archive extraction, which had marked E. Callosomarginal artery — that was wrong.

Q178. A 71-year-old man has a large left MCA stroke. Which artery is most likely affected?

Correct answer: D. ICA

Large MCA-territory strokes are frequently embolic FROM ipsilateral internal carotid artery (ICA) atherosclerotic disease — a classic artery-to-artery embolism mechanism, making ICA the most likely upstream embolic source.

Q179. Which statement is true?

Correct answer: A. Stroke = occlusion or rupture of cerebral vessels

'Stroke = neurological injury from occlusion or rupture of cerebral vessels' is the correct standard definition — the one true statement here.

Q180. A pituitary tumor typically causes which visual defect?

Correct answer: B. Optic chiasm lesion

A pituitary tumor classically compresses the optic chiasm midline, causing bitemporal hemianopia.

Q181. What is the first color typically lost in color blindness?

Archive answer: B. Green

FLAGGED — the 'first color lost' in color blindness varies by source and depends heavily on the specific type (congenital red-green vs. acquired, protanopia vs. deuteranopia vs. tritanopia); I don't have strong confidence in a single definitive answer without more context from your specific lecture material.

■ *Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.*

Q182. A patient has contralateral sensory loss and ipsilateral CN III palsy. This is caused by occlusion of the:

Correct answer: C. PCA

Contralateral sensory loss (thalamic involvement) plus ipsilateral CN III palsy (midbrain involvement) is consistent with a PCA-territory stroke, since PCA supplies both the thalamus and midbrain structures.

Q183. Which statement about venous drainage is correct?

Correct answer: B. Superior cerebral veins → superior sagittal sinus

Superior cerebral veins drain into the superior sagittal sinus — the one true statement among these options.

Q184. An ischemic stroke in the pontine cistern involves which artery?

Correct answer: A. Basilar artery

A pontine cistern lesion implicates the basilar artery.

Q185. A patient has an ischemic stroke in the pontine cistern. Which artery is likely to be occluded?

Correct answer: A. Basilar artery

Same fact as Q184/Q161/Q176 — the pontine cistern is basilar artery territory, not PCA (the archive's answer), which is inconsistent with this question's own repeated pattern elsewhere in the same archive.

✥ *Corrected from the original archive extraction, which had marked D. Posterior cerebral artery — that was wrong.*

Q186. Which of the following is NOT a criterion used in the ABCD2 score for stroke risk after a TIA?

Correct answer: A. Hyperlipidemia

The ABCD2 score components are Age, Blood pressure, Clinical features, Duration, and Diabetes — hyperlipidemia is a general stroke risk factor but not one of these five specific criteria.

Q187. An embolus from the internal carotid artery is most likely to lodge in which branch?

Correct answer: A. Middle cerebral artery (MCA)

MCA is the more direct continuation of ICA flow, making it the most common site for embolic lodgment from the internal carotid artery.

Q188. A patient has an embolus from the internal carotid artery. In which branch is the embolus most likely to lodge?

Correct answer: A. MCA

Same fact as Q187 — MCA.

Q189. The arterial supply of the primary visual cortex is the:

Correct answer: C. Calcarine artery (PCA branch)

The calcarine artery (a PCA branch) supplies the primary visual cortex.

Lecture 29 — Clinical Correlations (1 questions)

Q190. A 37-year-old woman presents with fatigue, difficulty speaking, ptosis, and diplopia that worsens throughout the day. Sensory...

Correct answer: E. Myasthenia Gravis

Fatigue, dysarthria, ptosis, and diplopia that WORSEN through the day, with a normal sensory exam, is the classic myasthenia gravis presentation.

Mixed / Other Topics (35 questions)

Q191. A 50-year-old man has ptosis, anhidrosis, and miosis due to a lung apex tumor. Which muscle is linked to the drooping eyelid?

Correct answer: D. Superior tarsal muscle

The ptosis (drooping eyelid) of Horner's syndrome results from paralysis of the superior tarsal (Müller's) muscle, a smooth muscle with sympathetic innervation that provides accessory lid elevation. The pupillary dilator (the archive's answer) relates instead to the MIOSIS finding, a different symptom.

✥ Corrected from the original archive extraction, which had marked C. Pupillary dilator — that was wrong.

Q192. A 50-year-old man with a lung apex tumor presents with miosis. Which muscle correlates with this finding?

Correct answer: C. Pupillary dilator

Miosis in Horner's syndrome results from loss of sympathetic tone to the pupillary dilator muscle, not arrector pili (the archive's answer, which relates to the anhidrosis/piloerection component instead, not pupil size).

✥ Corrected from the original archive extraction, which had marked B. Arrector pili — that was wrong.

Q193. A 40-year-old man has presbyopia (difficulty reading). Which statement about near focusing is correct?

Correct answer: C. M3 receptors contract ciliary muscle

M3 muscarinic (parasympathetic) receptors mediate ciliary muscle contraction for accommodation — a true statement. The archive answer is false: the pupillary sphincter is controlled by parasympathetic M3 receptors, not sympathetic alpha-2 receptors (alpha receptors instead control the dilator muscle).

✥ Corrected from the original archive extraction, which had marked D. α2 receptors contract sphincter muscle — that was wrong.

Q194. A 30-year-old woman has a UTI. Which receptor contracts the detrusor muscle?

Correct answer: E. M 3

Detrusor muscle contraction (for voiding) is mediated by parasympathetic M3 muscarinic receptors acting directly on bladder smooth muscle — not nicotinic (N) receptors, the archive's answer, which mediate the ganglionic synapse, not the final effector action.

✥ Corrected from the original archive extraction, which had marked C. N — that was wrong.

Q195. Which of the following is a sympathetic effect?

Correct answer: B. Increased heart rate

Increased heart rate is a classic sympathetic effect; the other options all describe parasympathetic effects instead.

Q196. Which receptor mediates detrusor muscle contraction for urination?

Correct answer: E. M3

Same fact as Q194 — M3 receptors mediate detrusor contraction. Beta-3 (the archive's answer) actually RELAXES the detrusor (for bladder storage), the opposite function.

✥ Corrected from the original archive extraction, which had marked B. β3 — that was wrong.

Q197. What is the best diagnostic test for Huntington’s disease?

Correct answer: C. Genetic testing

Huntington's is a single-gene disorder with definitive genetic testing available.

Q198. Huntington’s disease involves the loss of which neurotransmitter?

Correct answer: A. GABA

Huntington's involves loss of GABAergic medium spiny neurons in the striatum.

Q199. Which sympathetic receptor is used to contract the internal urinary sphincter?

Correct answer: A. Alpha-1

The internal urethral sphincter is contracted by sympathetic alpha-1 receptors (for continence/storage). Beta-3 (the archive's answer) instead relaxes the detrusor, a different structure/mechanism entirely.

✦ *Corrected from the original archive extraction, which had marked B. Beta-3 — that was wrong.*

Q200. What is the maximum GCS score?

Correct answer: C. 15

The maximum GCS score is 15 (4 eye + 5 verbal + 6 motor) — extremely well-established, basic clinical knowledge. '10' (the archive's answer) is simply incorrect.

✦ *Corrected from the original archive extraction, which had marked A. 10 — that was wrong.*

Q201. The nerve roots for the flexors of the arm (e.g., biceps) are:

Correct answer: A. C5–C6

Biceps reflex/arm flexor innervation is C5-C6.

Q202. The brain waves in an awake, alert state with eyes open are:

Correct answer: B. Beta

Awake, alert, EYES OPEN specifically points to beta rhythm — alpha rhythm is specifically associated with relaxed wakefulness with EYES CLOSED, and 'alpha blocking' shifts the pattern to beta once the eyes open.

Q203. A left corticobulbar lesion would result in which finding?

Correct answer: C. Weakness closing mouth tightly (right)

A left corticobulbar (UMN) lesion causes right lower-face weakness (e.g., can't close the mouth tightly), sparing upper-face/eyelid closure due to bilateral cortical innervation — confirms the same pattern established in Q76.

Q204. A patient has difficulty breathing and weakness in facial expression. Which part of the internal capsule might be affected?

Archive answer: D. Sublenticular part

FLAGGED — facial expression weakness points toward corticobulbar fibers, classically concentrated in the genu of the internal capsule. 'Sublenticular part' (the archive's answer) is more associated with auditory radiation fibers, not corticobulbar/facial pathways — moderate confidence this should be the genu instead, though the breathing-difficulty component of this question adds some uncertainty about the exact intended pathway.

■ *Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.*

Q205. What type of fiber provides input to the neostriatum?

Correct answer: B. Corticostriatal, excitatory via Glutamate

Illegible in the original archive. The corticostriatal pathway (cortex → striatum) is excitatory via glutamate — a major, well-established input to the neostriatum. The nigrostriatal pathway actually uses dopamine (not GABA or glutamate as the other options claim), and the thalamostriatal pathway is itself excitatory (glutamatergic), not inhibitory GABA as option D claims.

✦ *Corrected from the original archive extraction, which had marked None. (unmarked/illegible) — that was wrong.*

Q206. A patient presents with vertical diplopia and has downward gaze in the left eyeball when she adducts her eyes. What is the...

Correct answer: A. Left CN IV

Trouble looking down specifically when the eye is adducted, with vertical diplopia, is the classic CN IV (trochlear, superior oblique) palsy presentation.

Q207. What is the primary receptor for detrusor muscle contraction during urination?

Correct answer: E. M3

M3 muscarinic receptors mediate detrusor contraction for urination — confirms the correct answer for the same fact tested in Q194/Q196.

Q208. A patient comes with vertical diplopia and has difficulty looking down with the left eye, especially when the eye is adducted....

Correct answer: A. Left CN IV

Same CN IV palsy presentation as Q206.

Q209. The ganglia of the first-order taste neurons are located in which of the following structures?

Correct answer: A. Geniculate, Petrosal, Nodose ganglia

Geniculate (CN VII), petrosal (CN IX), and nodose (CN X) ganglia house the first-order taste neurons.

Q210. An unpleasant or distorted taste sensation is called:

Correct answer: C. Dysgeusia

Dysgeusia is the term for distorted or unpleasant taste.

Q211. When stereocilia of a hair cell bend toward the kinocilium:

Correct answer: C. Depolarization occurs

Stereocilia bending toward the kinocilium causes depolarization.

Q212. In the Gate-control theory of pain, enkephalins act by:

Correct answer: C. Inhibiting Ca2+ influx and opening K+ channels

Enkephalins work by inhibiting presynaptic Ca2+ influx AND opening postsynaptic K+ channels — reducing substance P release and dampening pain transmission. The archive answer 'Blocks Na+ channels presynaptically' doesn't describe the actual enkephalin/opioid receptor mechanism.

✥ Corrected from the original archive extraction, which had marked A. Blocking Na+ channels presynaptically — that was wrong.

Q213. Which neurotransmitter-transporter pair is CORRECT?

Correct answer: C. Serotonin — VMAT / SERT

Serotonin correctly pairs with VMAT (vesicular) and SERT (reuptake). The archive answer ('Glycine — DAT/GAT') is wrong; glycine doesn't use the dopamine transporter (DAT) at all — it has its own specific transporters.

✥ Corrected from the original archive extraction, which had marked E. Glycine — DAT / GAT — that was wrong.

Q214. Long-Term Potentiation (LTP) requires all of the following EXCEPT:

Correct answer: E. Activation of GABA-A receptors

GABA-A receptor activation (inhibitory) is NOT part of the LTP mechanism, which is fundamentally about excitatory glutamatergic strengthening. The archive answer 'AMPA receptor insertion' is actually a genuinely REQUIRED step of LTP expression (more AMPA receptors = a stronger response), so it can't be the 'except' item.

✥ Corrected from the original archive extraction, which had marked B. Insertion of AMPA receptors into the membrane — that was wrong.

Q215. The mechanism of action for diazepam (a benzodiazepine) is:

Correct answer: C. Increases the frequency of Cl– channel opening at the GABA-A receptor

Benzodiazepines like diazepam increase the FREQUENCY of Cl- channel opening at the GABA-A receptor (only when GABA is present) — distinct from barbiturates, which increase duration.

Q216. Which of the following is NOT a treatment for myasthenia gravis?

Correct answer: E. Dopamine agonists

Dopamine agonists have no role in myasthenia gravis treatment (that's a Parkinson's disease therapy). Thymectomy (the archive's answer) IS a genuine, standard MG treatment, especially with thymic pathology.

✥ Corrected from the original archive extraction, which had marked D. Thymectomy — that was wrong.

Q217. The neurotransmitter primarily linked to mood regulation is:

Correct answer: C. Serotonin

Serotonin is the neurotransmitter classically linked to mood regulation (the basis of SSRIs for depression). Glycine (the archive's answer) is an inhibitory amino acid neurotransmitter mainly in the spinal cord/brainstem, not classically tied to mood.

✥ Corrected from the original archive extraction, which had marked E. Glycine — that was wrong.

Q218. A lesion of the left Meyer's loop (temporal optic radiation) results in: 36

Correct answer: B. Right superior quadrantanopia

A left Meyer's loop lesion causes a RIGHT SUPERIOR quadrantanopia ('pie in the sky') — the archive answer 'Right inferior' describes the mirror-image PARIETAL radiation pathway instead.

✥ Corrected from the original archive extraction, which had marked C. Right inferior quadrantanopia — that was wrong.

Q219. A midline lesion of the optic chiasm (e.g., from a pituitary tumor) causes:

Correct answer: D. Bitemporal hemianopia

A midline chiasm lesion (pituitary tumor) causes bitemporal hemianopia — a chiasmal, not post-chiasmal, finding. The archive answer 'Right homonymous hemianopia' describes a post-chiasmal (tract/radiation/cortex) lesion instead, an entirely different anatomical level.

✥ Corrected from the original archive extraction, which had marked A. Right homonymous hemianopia — that was wrong.

Q220. In the semicircular canal system, during which condition does afferent firing return to baseline?

Correct answer: C. Constant-velocity rotation

Semicircular canals detect ACCELERATION/deceleration (dynamic changes), not constant velocity — afferent firing returns to baseline specifically during constant-velocity rotation, since the canals cannot detect steady-state velocity. The archive answer 'Angular deceleration' is itself a dynamic change that WOULD generate a signal, just in the opposite direction from acceleration.

✥ Corrected from the original archive extraction, which had marked B. Angular deceleration — that was wrong.

Q221. The primary receptor for cold sensation is:

Correct answer: C. TRPM8

TRPM8 is the classic cold/menthol receptor.

Q222. Which nerve does NOT synapse in the spinal trigeminal nucleus?

Correct answer: E. CN XI

CN XI (Accessory) is purely motor and does NOT contribute sensory fibers to the spinal trigeminal nucleus. CN VII (the archive's answer) actually DOES contribute a small GSA component there, along with V, IX, and X.

✥ Corrected from the original archive extraction, which had marked B. CN VII — that was wrong.

Q223. The mechanism of action of fluoxetine is:

Correct answer: D. Inhibits SERT

Fluoxetine is an SSRI — it inhibits SERT (serotonin reuptake transporter). 'Enhances GABA binding' (the archive's answer) describes benzodiazepines, a completely different drug class.

✥ Corrected from the original archive extraction, which had marked E. Enhances GABA binding — that was wrong.

Q224. A lesion of the left dorsal (parietal) optic radiation produces:

Correct answer: B. Right inferior quadrantanopia

Parietal (dorsal) optic radiation fibers carry the contralateral INFERIOR visual field — a left parietal radiation lesion causes right inferior quadrantanopia ('pie on the floor'), the mirror image of Meyer's loop's superior-field territory. The archive answer 'Right superior' describes the temporal (Meyer's loop) pathway instead.

✥ Corrected from the original archive extraction, which had marked A. Right superior quadrantanopia — that was wrong.

Q225. Neurofibromatosis primarily affects which cell type?

Correct answer: C. Schwann cell

Neurofibromatosis classically involves Schwann cell tumors (neurofibromas, schwannomas).